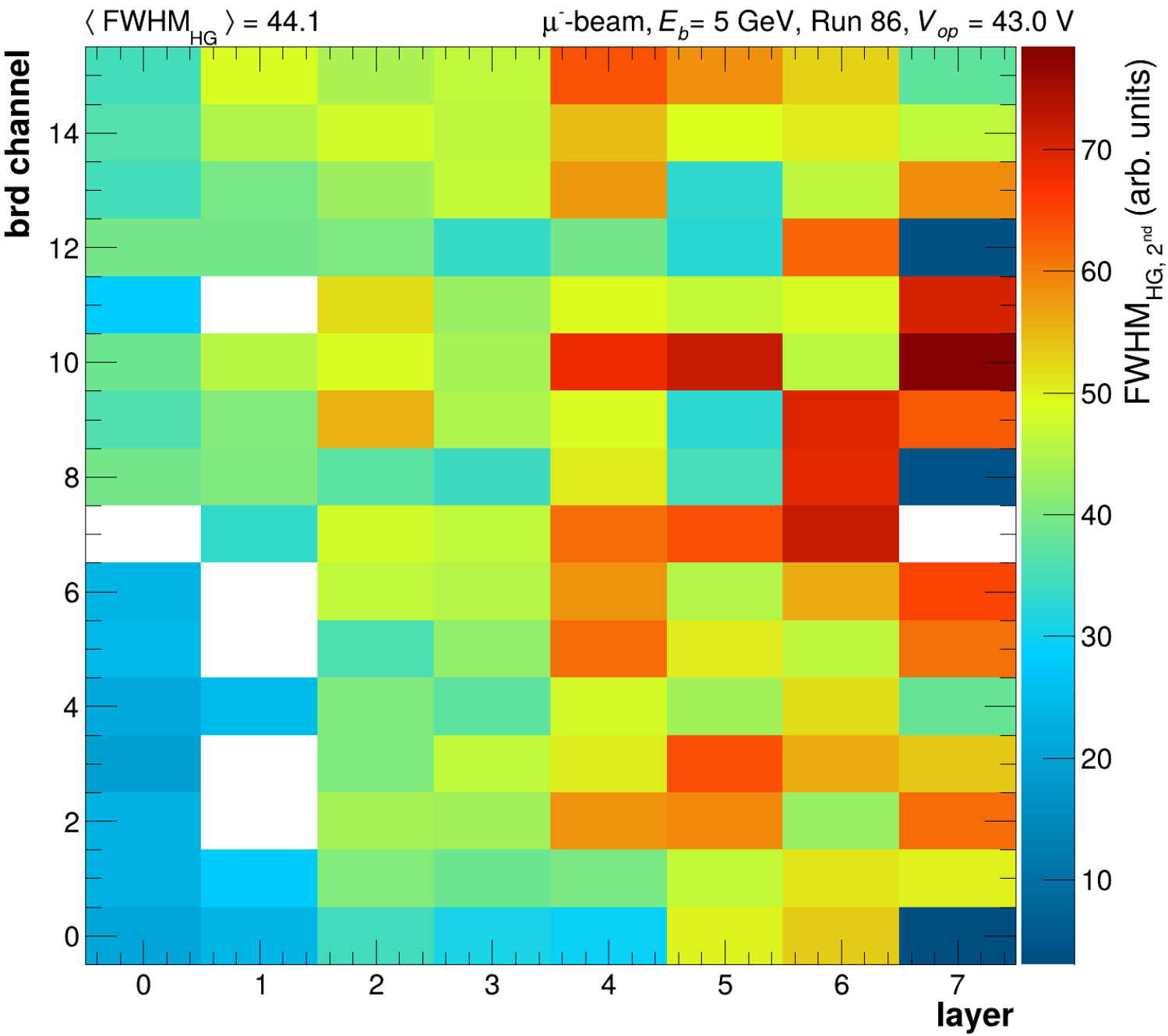
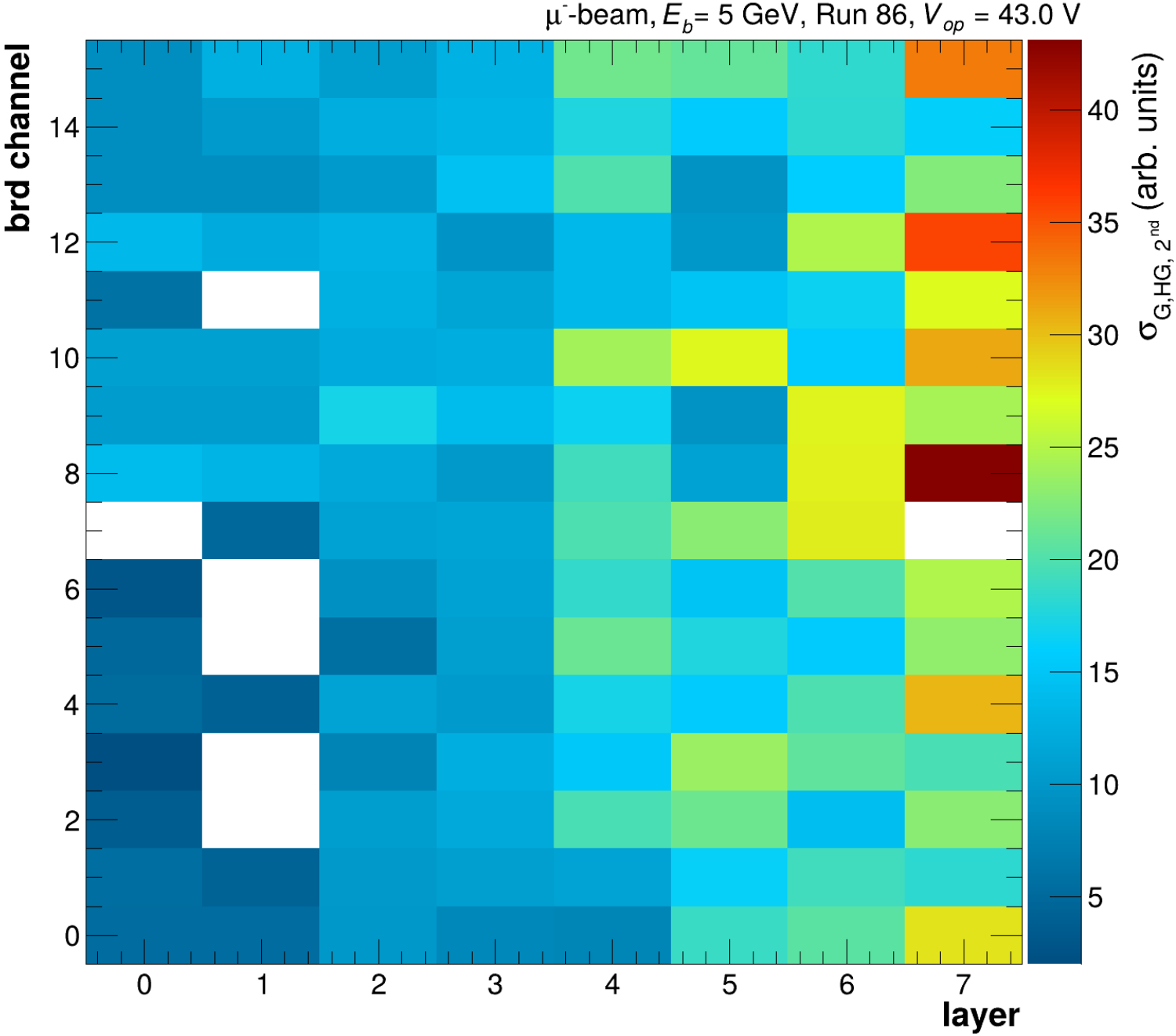
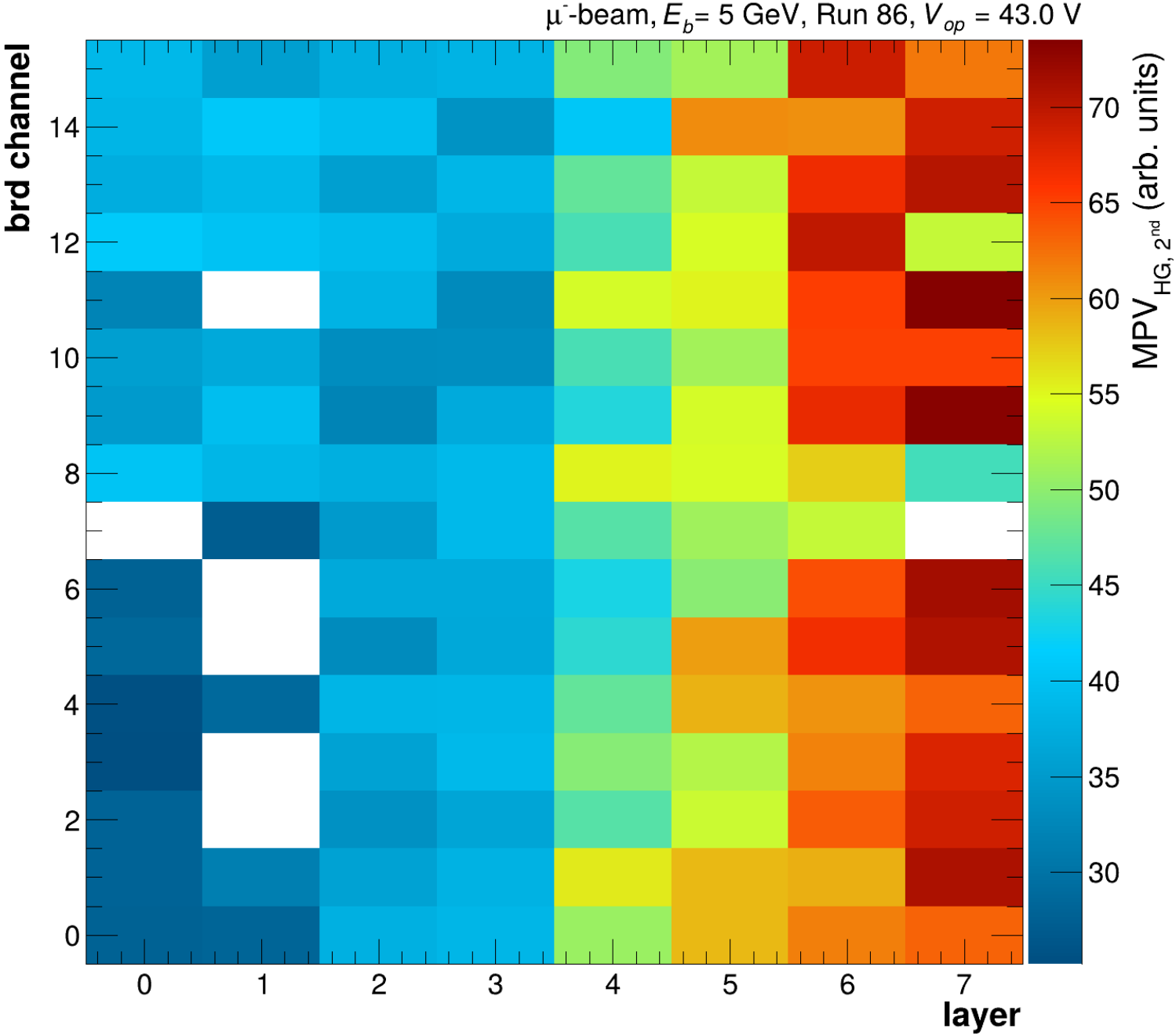
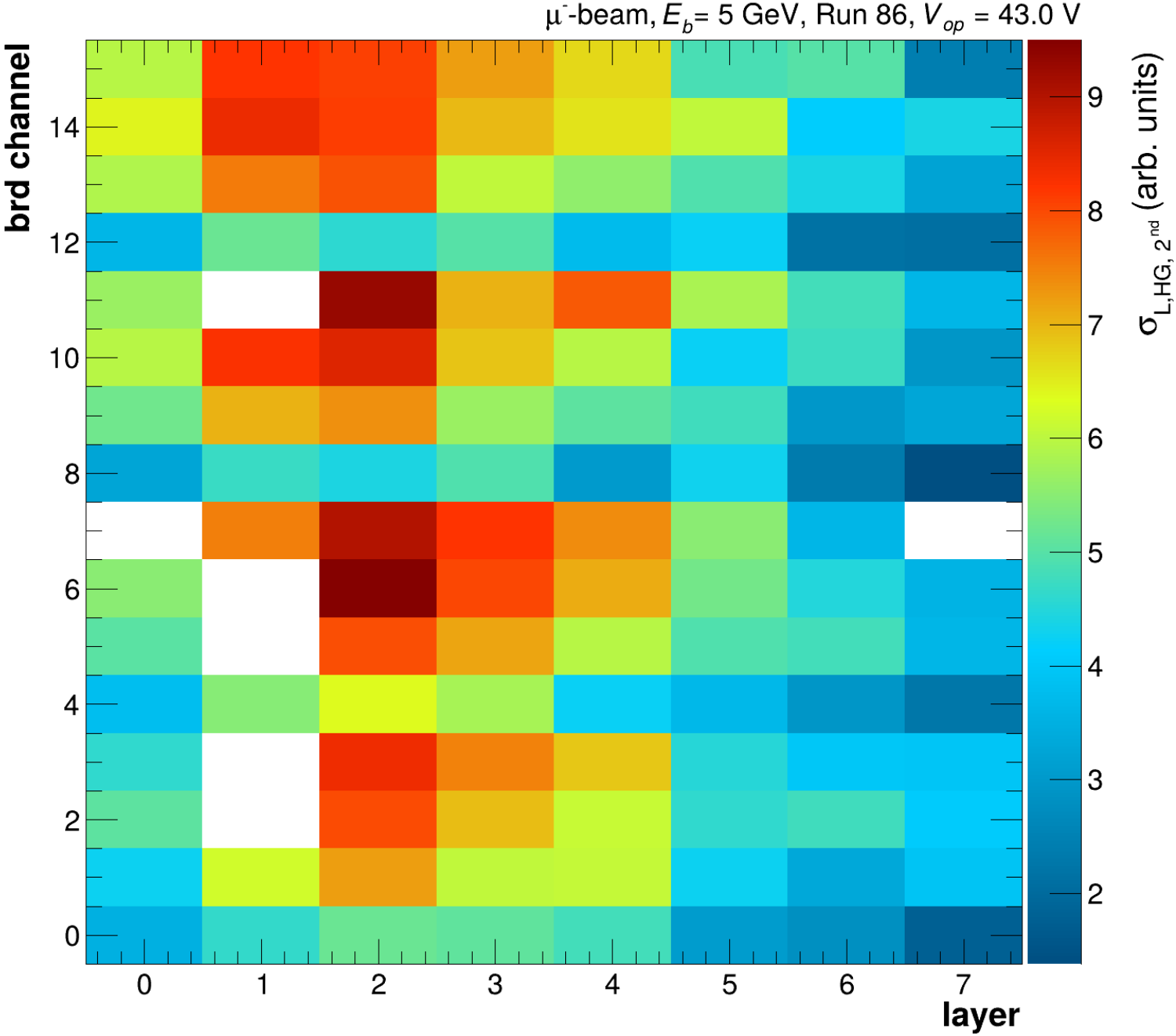
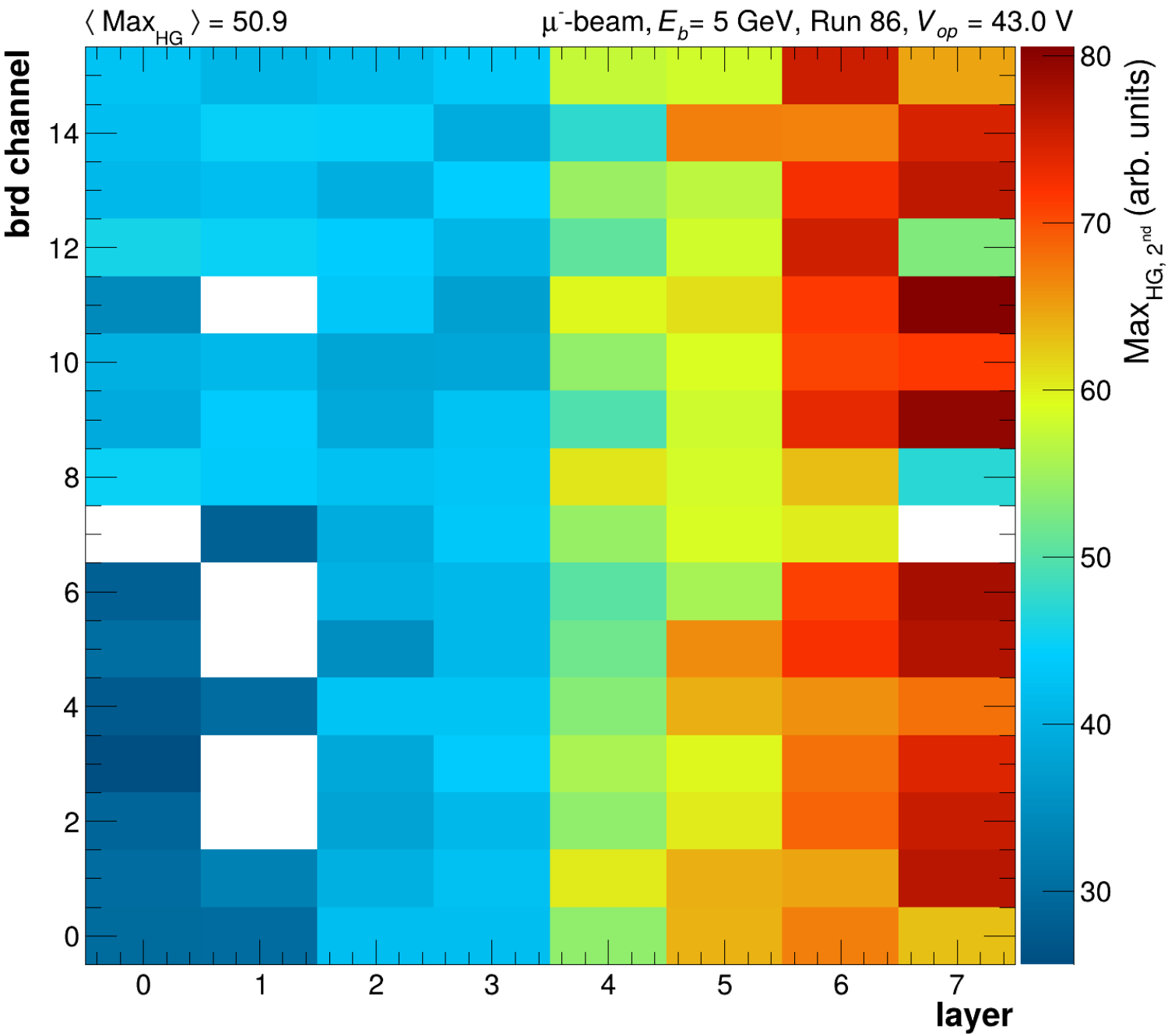
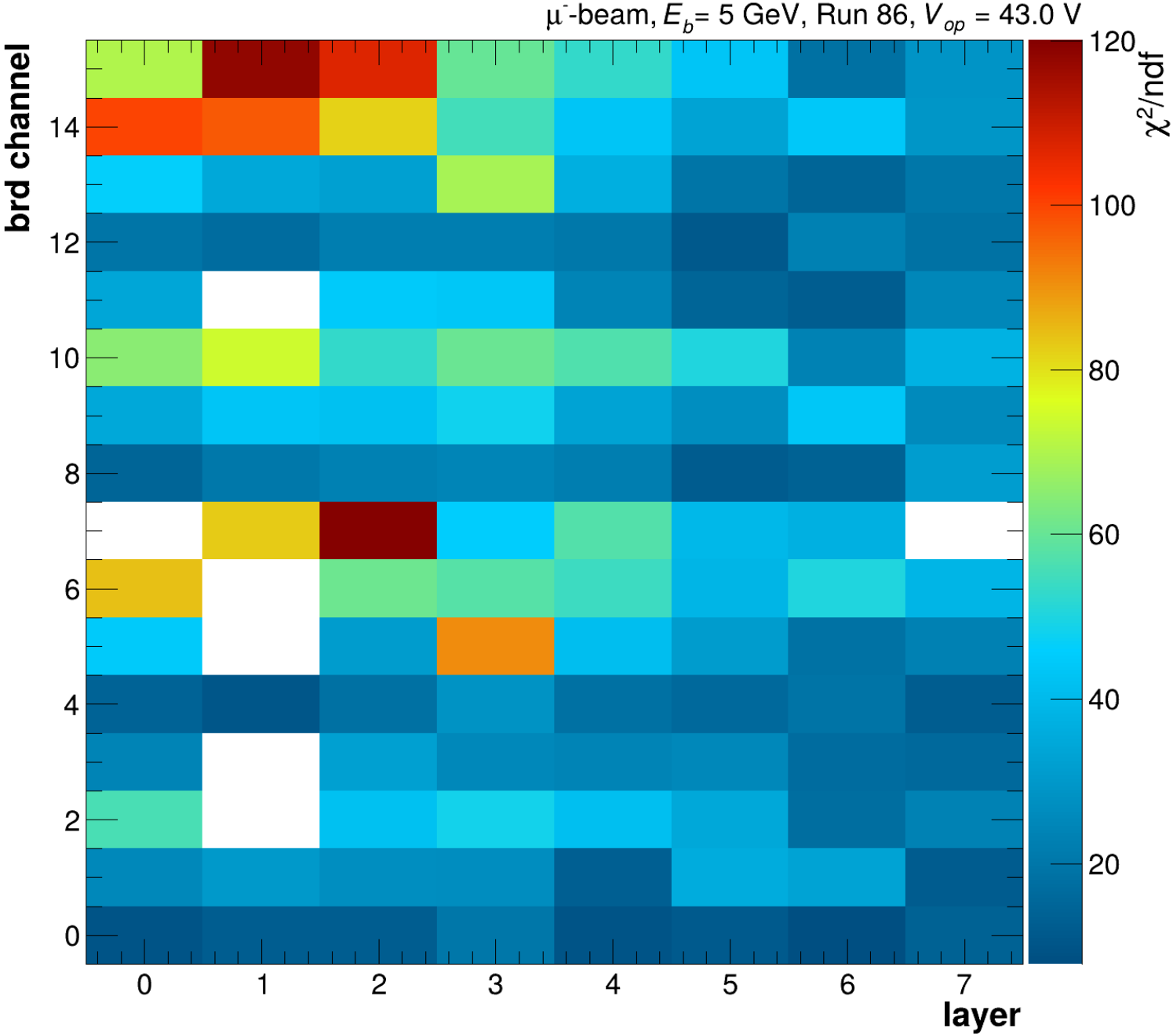

⟨ FWHM_HG ⟩ = 44.1
μ⁻-beam, E_b= 5 GeV, Run 86, V_op = 43.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60
70

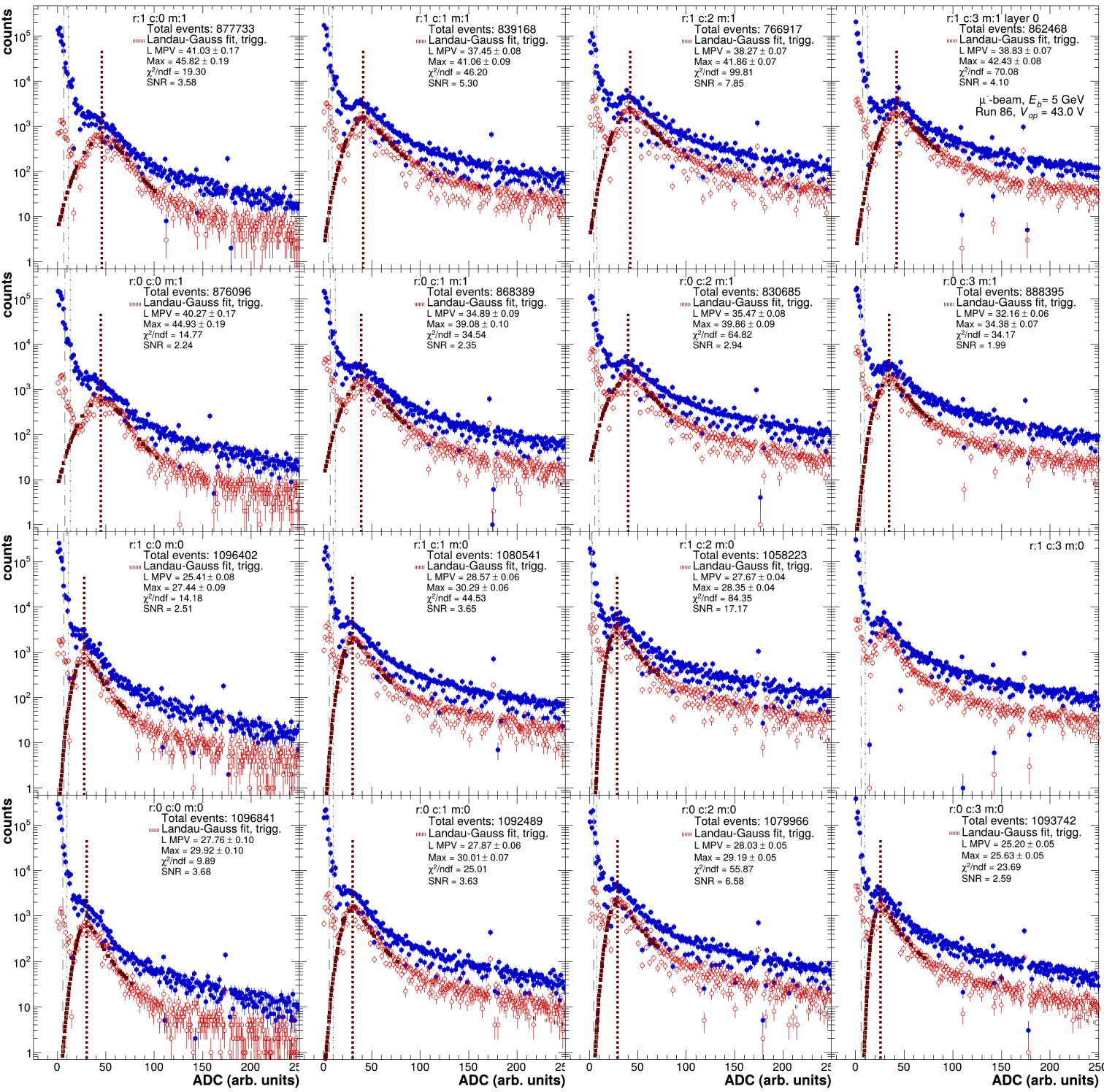

μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
5
10
15
20
25
30
35
40

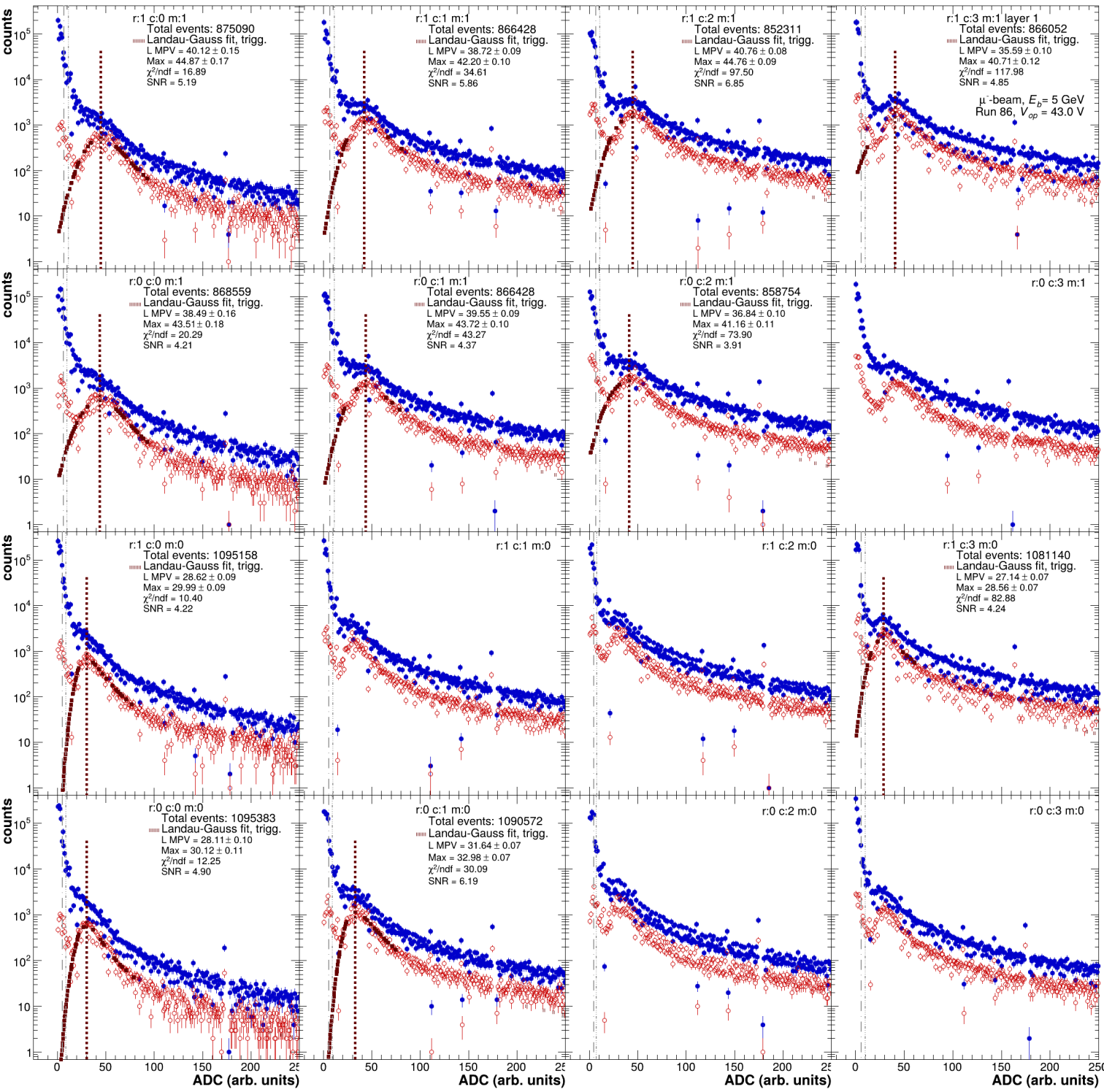

μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
30
35
40
45
50
55
60
65
70

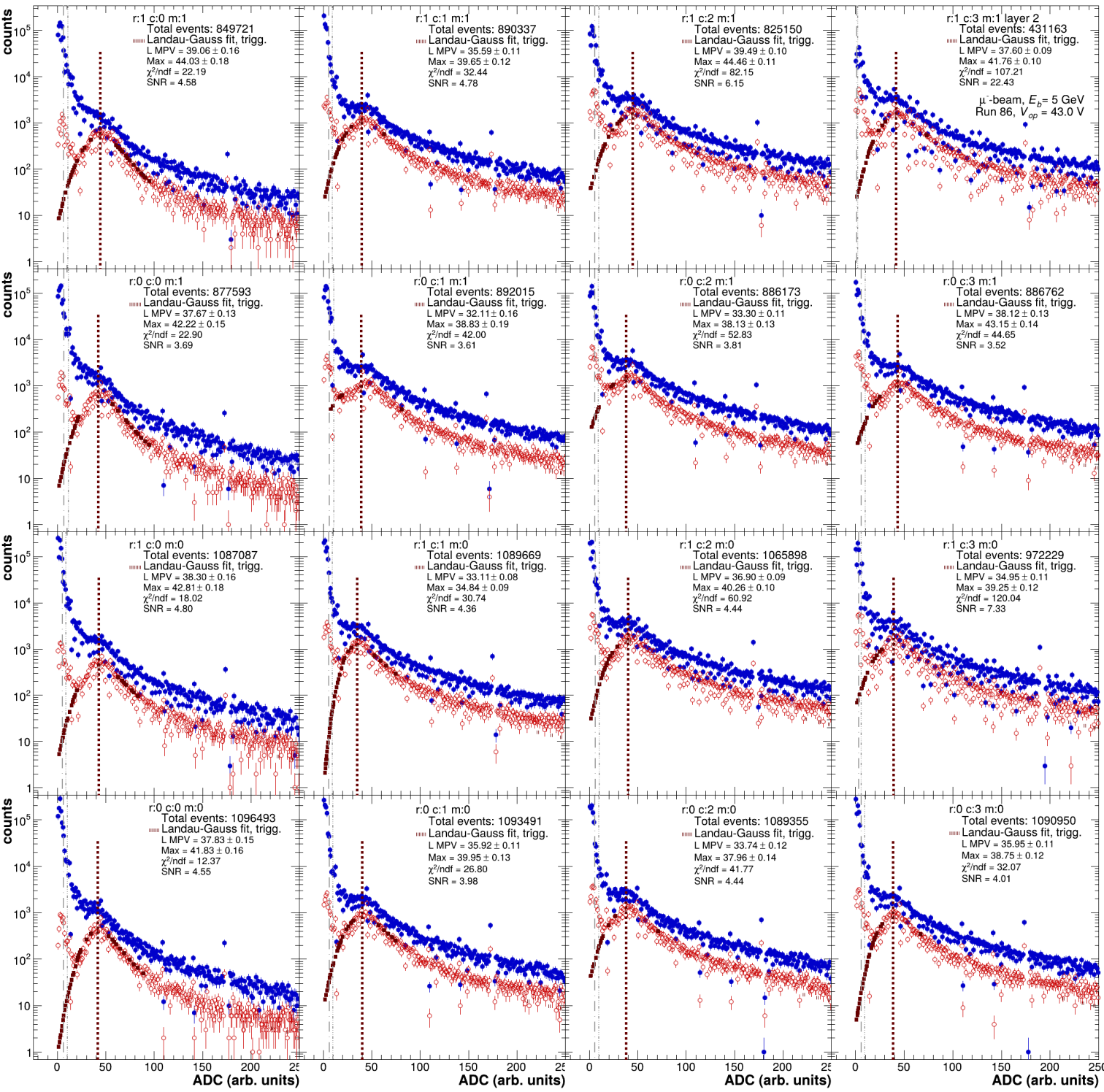

μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
2
3
4
5
6
7
8
9

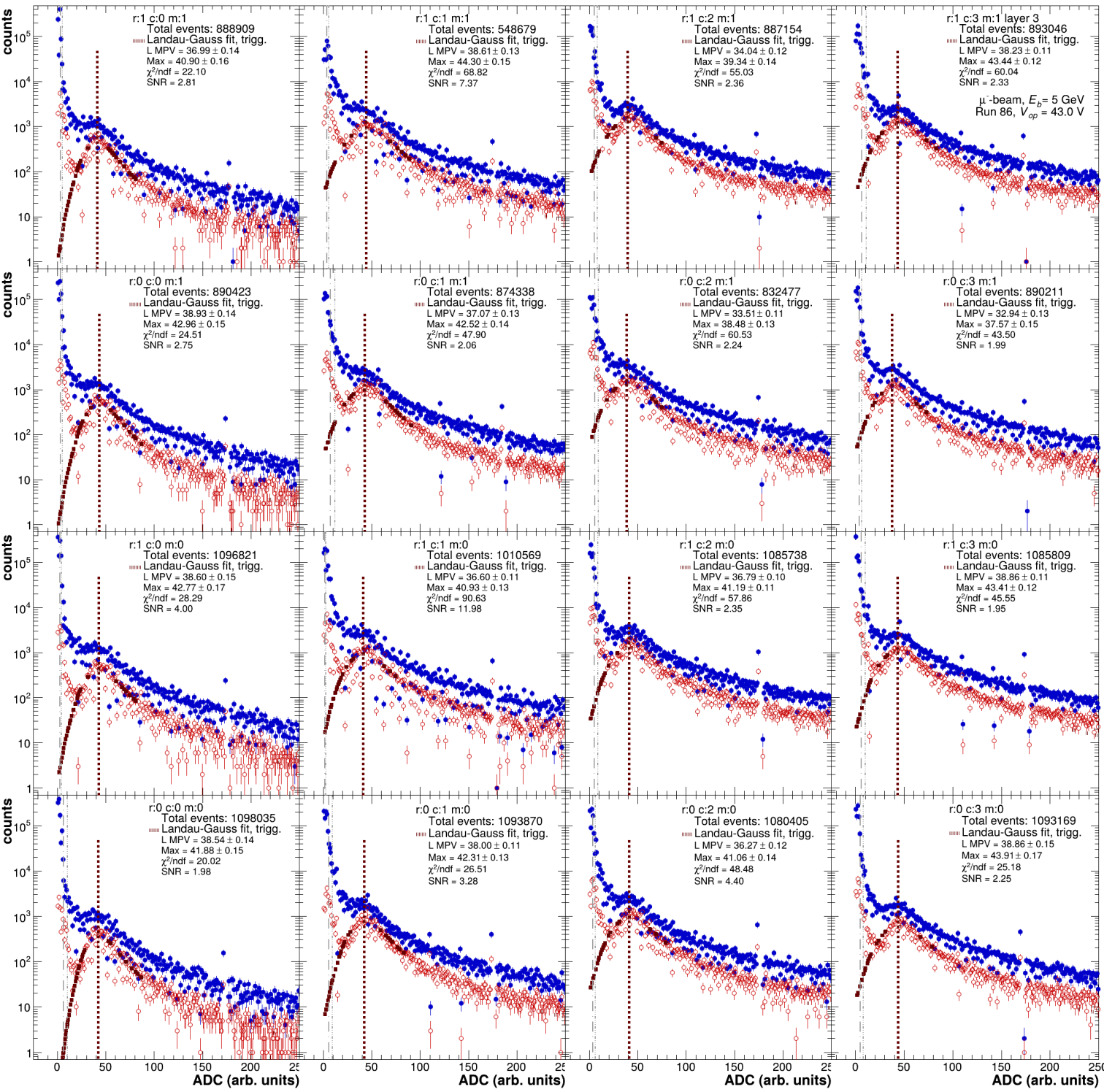

⟨ Max_HG ⟩ = 50.9
μ⁻-beam, E_b= 5 GeV, Run 86, V_op = 43.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
40
50
60
70
80

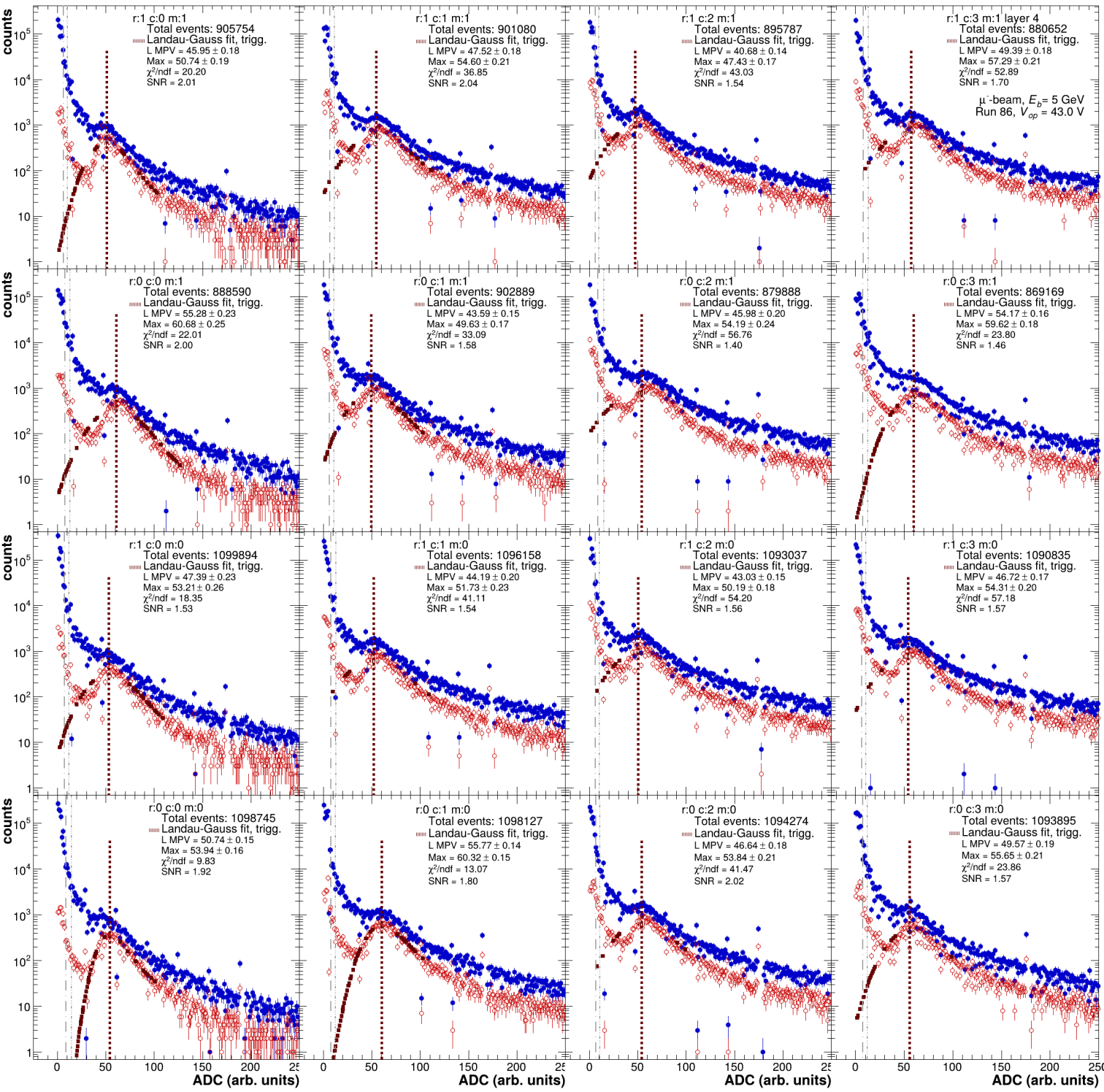

μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
120
100
80
60
40
20
χ^2/ndf

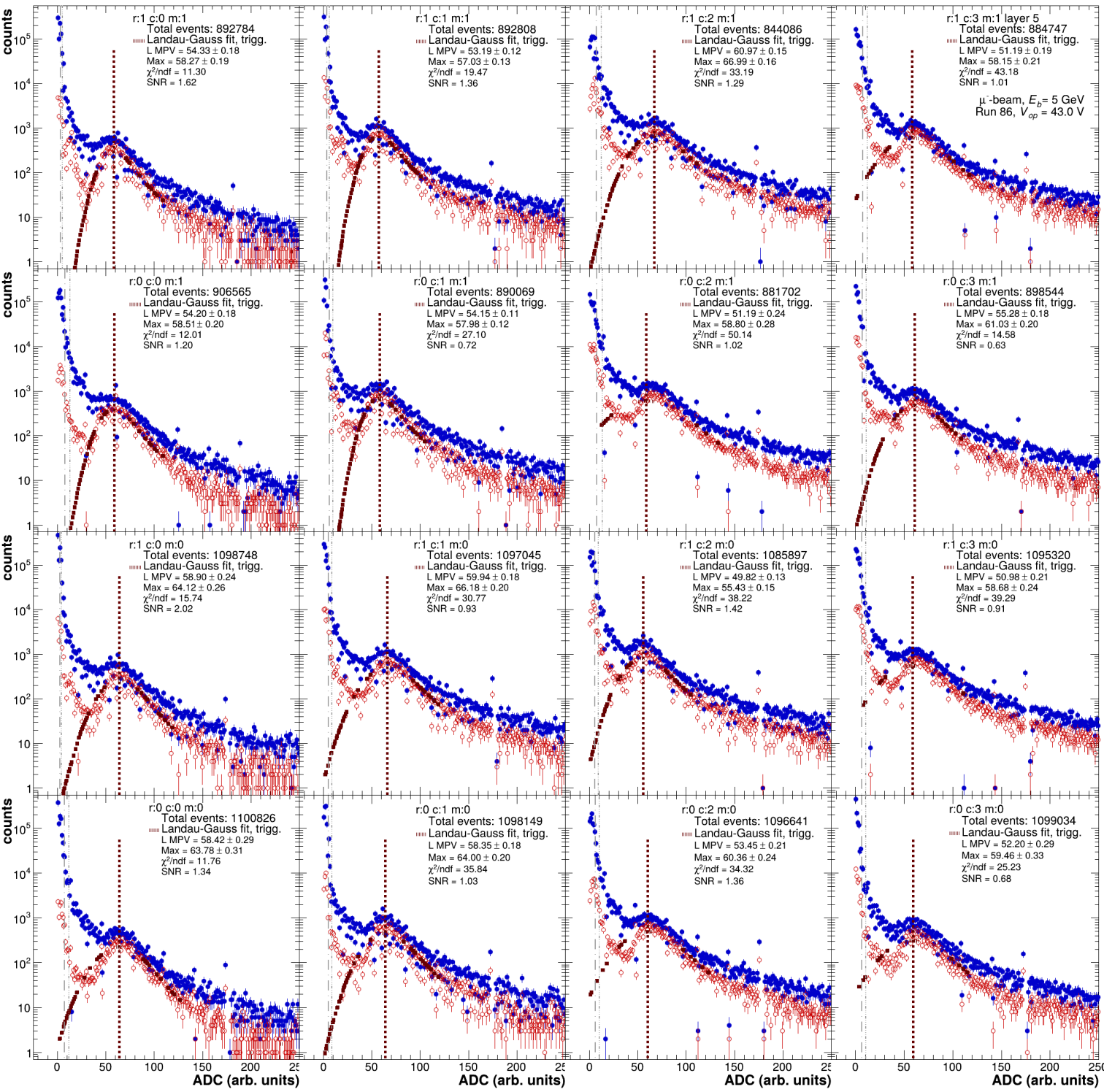

r:1 c:0 m:1
Total events: 877733
Landau-Gauss fit, trigg.
L MPV = 41.03 ± 0.17
Max = 45.82 ± 0.19
χ²/ndf = 19.30
SNR = 3.58
r:1 c:1 m:1
Total events: 839168
Landau-Gauss fit, trigg.
L MPV = 37.45 ± 0.08
Max = 41.06 ± 0.09
χ²/ndf = 46.20
SNR = 5.30
r:1 c:2 m:1
Total events: 766917
Landau-Gauss fit, trigg.
L MPV = 38.27 ± 0.07
Max = 41.86 ± 0.07
χ²/ndf = 99.81
SNR = 7.85
r:1 c:3 m:1 layer 0
Total events: 862468
Landau-Gauss fit, trigg.
L MPV = 38.83 ± 0.07
Max = 42.43 ± 0.08
χ²/ndf = 70.08
SNR = 4.10
μ⁻-beam, E_b = 5 GeV
Run 86, V_op = 43.0 V
r:0 c:0 m:1
Total events: 876096
Landau-Gauss fit, trigg.
L MPV = 40.27 ± 0.17
Max = 44.93 ± 0.19
χ²/ndf = 14.77
SNR = 2.24
r:0 c:1 m:1
Total events: 868389
Landau-Gauss fit, trigg.
L MPV = 34.89 ± 0.09
Max = 39.08 ± 0.10
χ²/ndf = 34.54
SNR = 2.35
r:0 c:2 m:1
Total events: 830685
Landau-Gauss fit, trigg.
L MPV = 35.47 ± 0.08
Max = 39.86 ± 0.09
χ²/ndf = 64.82
SNR = 2.94
r:0 c:3 m:1
Total events: 888395
Landau-Gauss fit, trigg.
L MPV = 32.16 ± 0.06
Max = 34.38 ± 0.07
χ²/ndf = 34.17
SNR = 1.99
r:1 c:0 m:0
Total events: 1096402
Landau-Gauss fit, trigg.
L MPV = 25.41 ± 0.08
Max = 27.44 ± 0.09
χ²/ndf = 14.18
SNR = 2.51
r:1 c:1 m:0
Total events: 1080541
Landau-Gauss fit, trigg.
L MPV = 28.57 ± 0.06
Max = 30.29 ± 0.06
χ²/ndf = 44.53
SNR = 3.65
r:1 c:2 m:0
Total events: 1058223
Landau-Gauss fit, trigg.
L MPV = 27.67 ± 0.04
Max = 28.35 ± 0.04
χ²/ndf = 84.35
SNR = 17.17
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 1096841
Landau-Gauss fit, trigg.
L MPV = 27.76 ± 0.10
Max = 29.92 ± 0.10
χ²/ndf = 9.89
SNR = 3.68
r:0 c:1 m:0
Total events: 1092489
Landau-Gauss fit, trigg.
L MPV = 27.87 ± 0.06
Max = 30.01 ± 0.07
χ²/ndf = 25.01
SNR = 3.63
r:0 c:2 m:0
Total events: 1079966
Landau-Gauss fit, trigg.
L MPV = 28.03 ± 0.05
Max = 29.19 ± 0.05
χ²/ndf = 55.87
SNR = 6.58
r:0 c:3 m:0
Total events: 1093742
Landau-Gauss fit, trigg.
L MPV = 25.20 ± 0.05
Max = 25.63 ± 0.05
χ²/ndf = 23.69
SNR = 2.59
counts
ADC (arb. units)

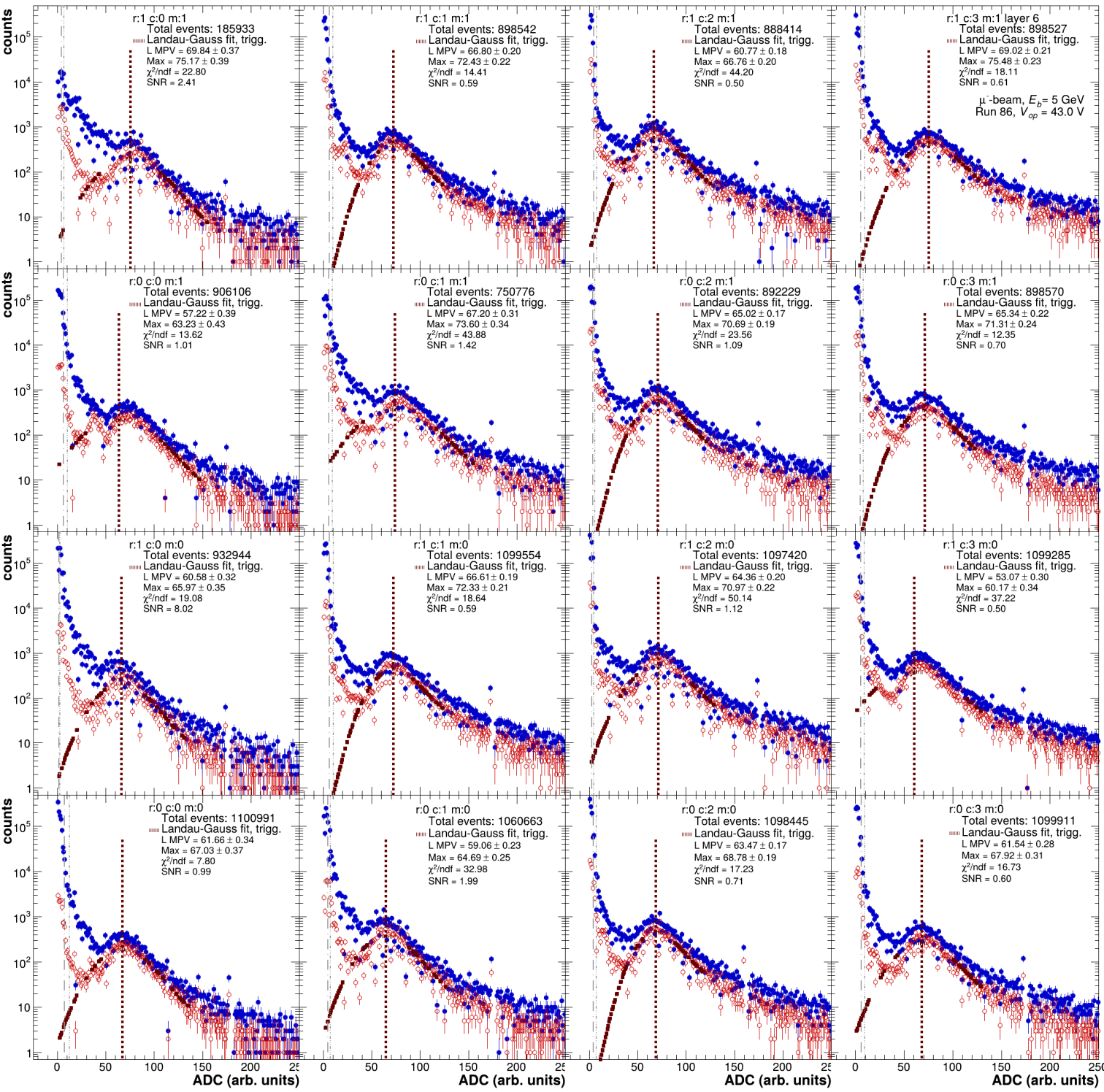

r:1 c:0 m:1
Total events: 875090
Landau-Gauss fit, trigg.
L MPV = 40.12 ± 0.15
Max = 44.87 ± 0.17
χ²/ndf = 16.89
SNR = 5.19
r:1 c:1 m:1
Total events: 866428
Landau-Gauss fit, trigg.
L MPV = 38.72 ± 0.09
Max = 42.20 ± 0.10
χ²/ndf = 34.61
SNR = 5.86
r:1 c:2 m:1
Total events: 852311
Landau-Gauss fit, trigg.
L MPV = 40.76 ± 0.08
Max = 44.76 ± 0.09
χ²/ndf = 97.50
SNR = 6.85
r:1 c:3 m:1 layer 1
Total events: 866052
Landau-Gauss fit, trigg.
L MPV = 35.59 ± 0.10
Max = 40.71 ± 0.12
χ²/ndf = 117.98
SNR = 4.85
μ⁻-beam, E_b = 5 GeV
Run 86, V_op = 43.0 V
r:0 c:0 m:1
Total events: 868559
Landau-Gauss fit, trigg.
L MPV = 38.49 ± 0.16
Max = 43.51 ± 0.18
χ²/ndf = 20.29
SNR = 4.21
r:0 c:1 m:1
Total events: 866428
Landau-Gauss fit, trigg.
L MPV = 39.55 ± 0.09
Max = 43.72 ± 0.10
χ²/ndf = 43.27
SNR = 4.37
r:0 c:2 m:1
Total events: 858754
Landau-Gauss fit, trigg.
L MPV = 36.84 ± 0.10
Max = 41.16 ± 0.11
χ²/ndf = 73.90
SNR = 3.91
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 1095158
Landau-Gauss fit, trigg.
L MPV = 28.62 ± 0.09
Max = 29.99 ± 0.09
χ²/ndf = 10.40
SNR = 4.22
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
Total events: 1081140
Landau-Gauss fit, trigg.
L MPV = 27.14 ± 0.07
Max = 28.56 ± 0.07
χ²/ndf = 82.88
SNR = 4.24
r:0 c:0 m:0
Total events: 1095383
Landau-Gauss fit, trigg.
L MPV = 28.11 ± 0.10
Max = 30.12 ± 0.11
χ²/ndf = 12.25
SNR = 4.90
r:0 c:1 m:0
Total events: 1090572
Landau-Gauss fit, trigg.
L MPV = 31.64 ± 0.07
Max = 32.98 ± 0.07
χ²/ndf = 30.09
SNR = 6.19
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

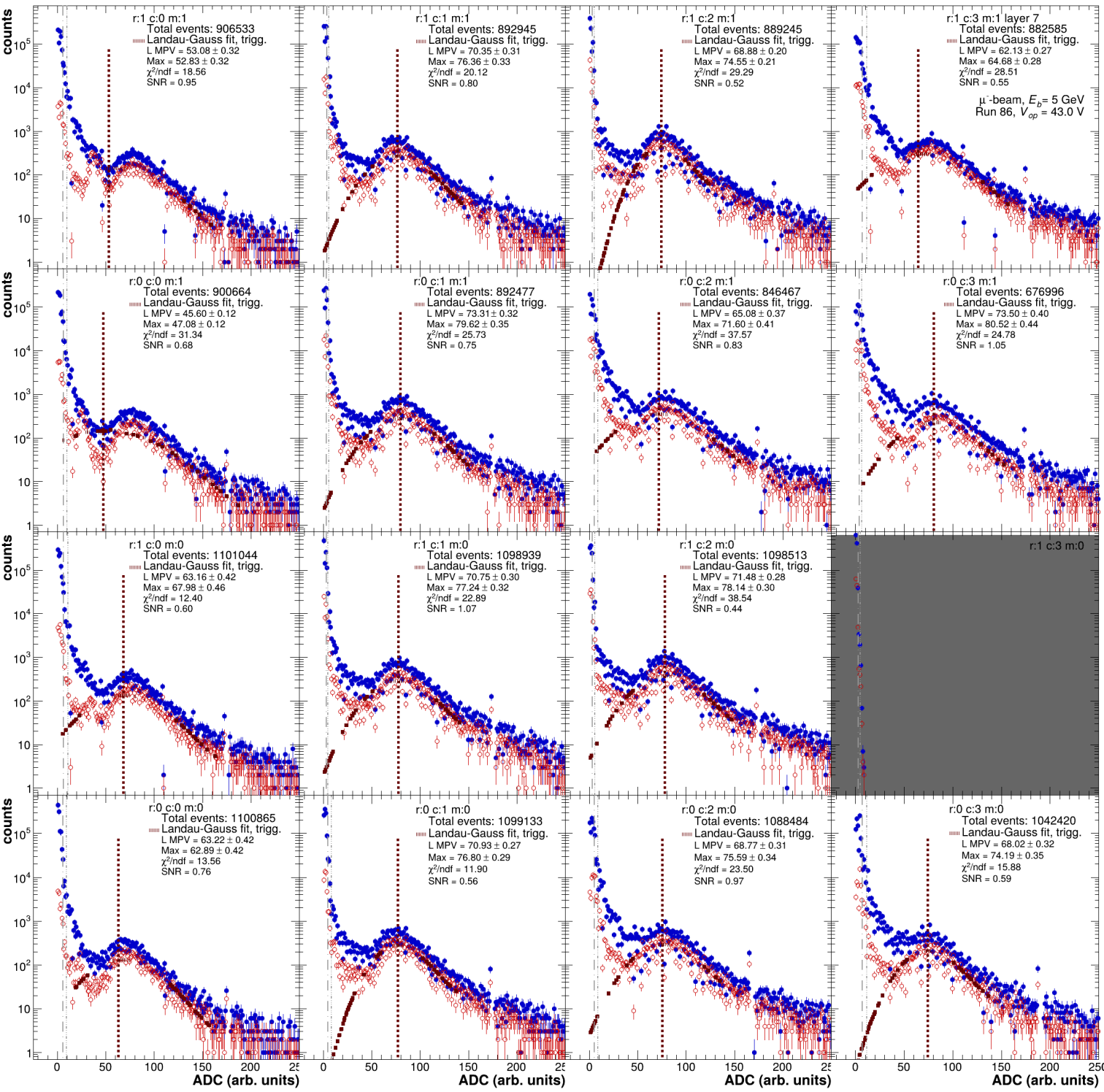

r:1 c:0 m:1
Total events: 849721
Landau-Gauss fit, trigg.
L MPV = 39.06 ± 0.16
Max = 44.03 ± 0.18
χ²/ndf = 22.19
SNR = 4.58
r:1 c:1 m:1
Total events: 890337
Landau-Gauss fit, trigg.
L MPV = 35.59 ± 0.11
Max = 39.65 ± 0.12
χ²/ndf = 32.44
SNR = 4.78
r:1 c:2 m:1
Total events: 825150
Landau-Gauss fit, trigg.
L MPV = 39.49 ± 0.10
Max = 44.46 ± 0.11
χ²/ndf = 82.15
SNR = 6.15
r:1 c:3 m:1 layer 2
Total events: 431163
Landau-Gauss fit, trigg.
L MPV = 37.60 ± 0.09
Max = 41.76 ± 0.10
χ²/ndf = 107.21
SNR = 22.43
μ⁻-beam, E_b = 5 GeV
Run 86, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 877593
Landau-Gauss fit, trigg.
L MPV = 37.67 ± 0.13
Max = 42.22 ± 0.15
χ²/ndf = 22.90
SNR = 3.69
r:0 c:1 m:1
Total events: 892015
Landau-Gauss fit, trigg.
L MPV = 32.11 ± 0.16
Max = 38.83 ± 0.19
χ²/ndf = 42.00
SNR = 3.61
r:0 c:2 m:1
Total events: 886173
Landau-Gauss fit, trigg.
L MPV = 33.30 ± 0.11
Max = 38.13 ± 0.13
χ²/ndf = 52.83
SNR = 3.81
r:0 c:3 m:1
Total events: 886762
Landau-Gauss fit, trigg.
L MPV = 38.12 ± 0.13
Max = 43.15 ± 0.14
χ²/ndf = 44.65
SNR = 3.52
r:1 c:0 m:0
Total events: 1087087
Landau-Gauss fit, trigg.
L MPV = 38.30 ± 0.16
Max = 42.81 ± 0.18
χ²/ndf = 18.02
SNR = 4.80
r:1 c:1 m:0
Total events: 1089669
Landau-Gauss fit, trigg.
L MPV = 33.11 ± 0.08
Max = 34.84 ± 0.09
χ²/ndf = 30.74
SNR = 4.36
r:1 c:2 m:0
Total events: 1065898
Landau-Gauss fit, trigg.
L MPV = 36.90 ± 0.09
Max = 40.26 ± 0.10
χ²/ndf = 60.92
SNR = 4.44
r:1 c:3 m:0
Total events: 972229
Landau-Gauss fit, trigg.
L MPV = 34.95 ± 0.11
Max = 39.25 ± 0.12
χ²/ndf = 120.04
SNR = 7.33
r:0 c:0 m:0
Total events: 1096493
Landau-Gauss fit, trigg.
L MPV = 37.83 ± 0.15
Max = 41.83 ± 0.16
χ²/ndf = 12.37
SNR = 4.55
r:0 c:1 m:0
Total events: 1093491
Landau-Gauss fit, trigg.
L MPV = 35.92 ± 0.11
Max = 39.95 ± 0.13
χ²/ndf = 26.80
SNR = 3.98
r:0 c:2 m:0
Total events: 1089355
Landau-Gauss fit, trigg.
L MPV = 33.74 ± 0.12
Max = 37.96 ± 0.14
χ²/ndf = 41.77
SNR = 4.44
r:0 c:3 m:0
Total events: 1090950
Landau-Gauss fit, trigg.
L MPV = 35.95 ± 0.11
Max = 38.75 ± 0.12
χ²/ndf = 32.07
SNR = 4.01
counts
ADC (arb. units)

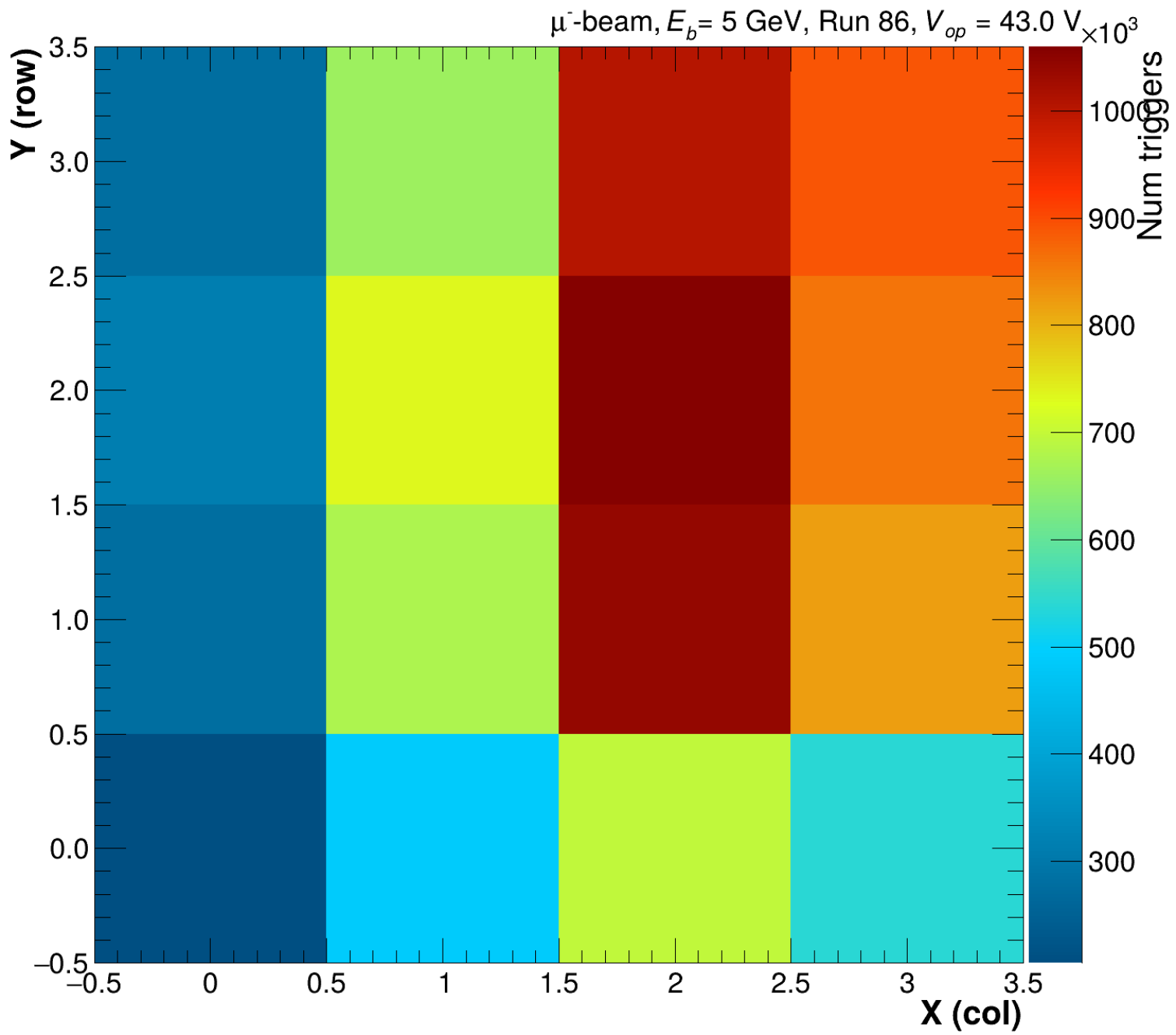

r:1 c:0 m:1
Total events: 888909
Landau-Gauss fit, trigg.
L MPV = 36.99 ± 0.14
Max = 40.90 ± 0.16
χ²/ndf = 22.10
SNR = 2.81
r:1 c:1 m:1
Total events: 548679
Landau-Gauss fit, trigg.
L MPV = 38.61 ± 0.13
Max = 44.30 ± 0.15
χ²/ndf = 68.82
SNR = 7.37
r:1 c:2 m:1
Total events: 887154
Landau-Gauss fit, trigg.
L MPV = 34.04 ± 0.12
Max = 39.34 ± 0.14
χ²/ndf = 55.03
SNR = 2.36
r:1 c:3 m:1 layer 3
Total events: 893046
Landau-Gauss fit, trigg.
L MPV = 38.23 ± 0.11
Max = 43.44 ± 0.12
χ²/ndf = 60.04
SNR = 2.33
μ⁻-beam, E_b = 5 GeV
Run 86, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 890423
Landau-Gauss fit, trigg.
L MPV = 38.93 ± 0.14
Max = 42.96 ± 0.15
χ²/ndf = 24.51
SNR = 2.75
r:0 c:1 m:1
Total events: 874338
Landau-Gauss fit, trigg.
L MPV = 37.07 ± 0.13
Max = 42.52 ± 0.14
χ²/ndf = 47.90
SNR = 2.06
r:0 c:2 m:1
Total events: 832477
Landau-Gauss fit, trigg.
L MPV = 33.51 ± 0.11
Max = 38.48 ± 0.13
χ²/ndf = 60.53
SNR = 2.24
r:0 c:3 m:1
Total events: 890211
Landau-Gauss fit, trigg.
L MPV = 32.94 ± 0.13
Max = 37.57 ± 0.15
χ²/ndf = 43.50
SNR = 1.99
r:1 c:0 m:0
Total events: 1096821
Landau-Gauss fit, trigg.
L MPV = 38.60 ± 0.15
Max = 42.77 ± 0.17
χ²/ndf = 28.29
SNR = 4.00
r:1 c:1 m:0
Total events: 1010569
Landau-Gauss fit, trigg.
L MPV = 36.60 ± 0.11
Max = 40.93 ± 0.13
χ²/ndf = 90.63
SNR = 11.98
r:1 c:2 m:0
Total events: 1085738
Landau-Gauss fit, trigg.
L MPV = 36.79 ± 0.10
Max = 41.19 ± 0.11
χ²/ndf = 57.86
SNR = 2.35
r:1 c:3 m:0
Total events: 1085809
Landau-Gauss fit, trigg.
L MPV = 38.86 ± 0.11
Max = 43.41 ± 0.12
χ²/ndf = 45.55
SNR = 1.95
r:0 c:0 m:0
Total events: 1098035
Landau-Gauss fit, trigg.
L MPV = 38.54 ± 0.14
Max = 41.88 ± 0.15
χ²/ndf = 20.02
SNR = 1.98
r:0 c:1 m:0
Total events: 1093870
Landau-Gauss fit, trigg.
L MPV = 38.00 ± 0.11
Max = 42.31 ± 0.13
χ²/ndf = 26.51
SNR = 3.28
r:0 c:2 m:0
Total events: 1080405
Landau-Gauss fit, trigg.
L MPV = 36.27 ± 0.12
Max = 41.06 ± 0.14
χ²/ndf = 48.48
SNR = 4.40
r:0 c:3 m:0
Total events: 1093169
Landau-Gauss fit, trigg.
L MPV = 38.86 ± 0.15
Max = 43.91 ± 0.17
χ²/ndf = 25.18
SNR = 2.25
counts
ADC (arb. units)

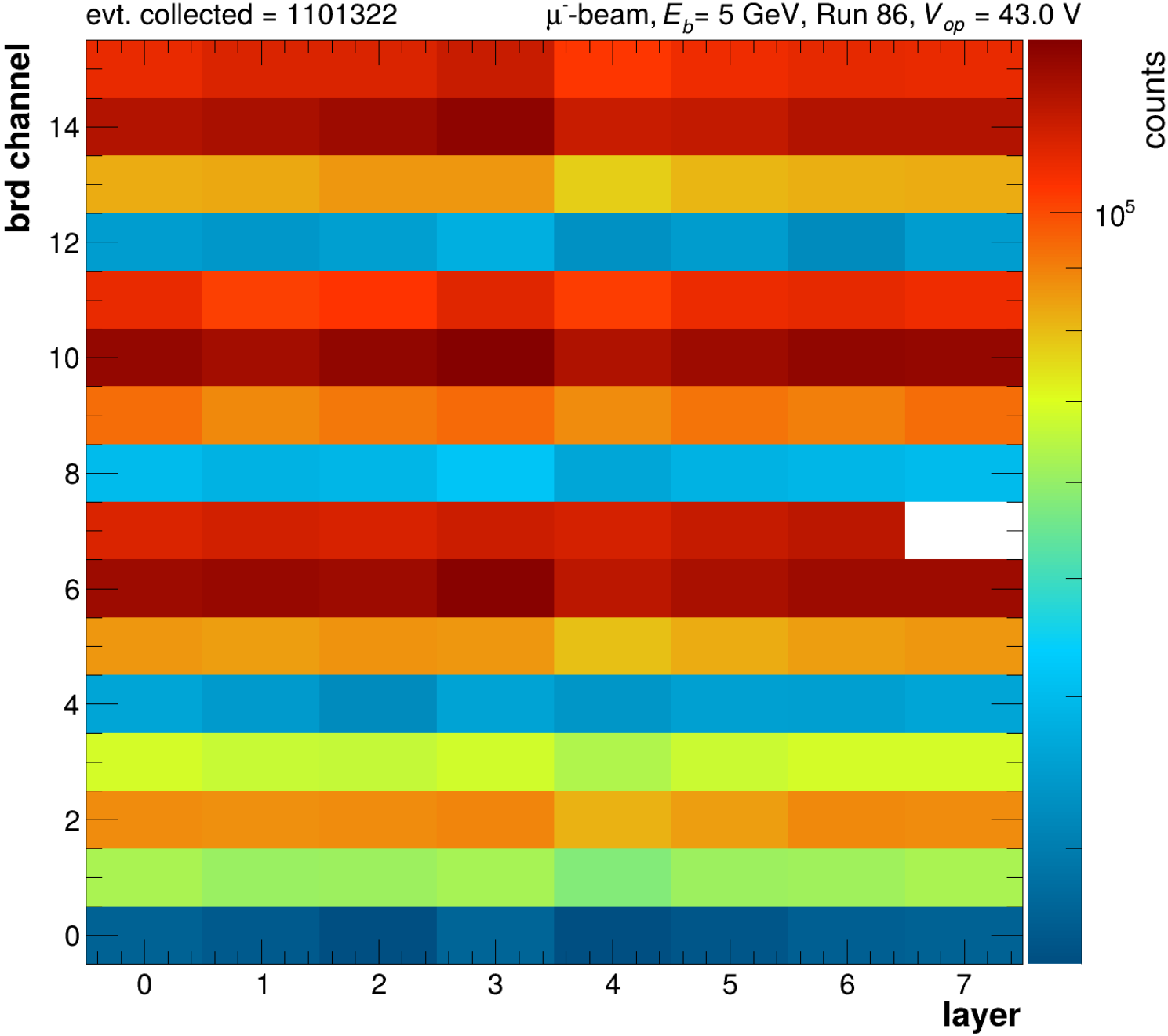

r:1 c:0 m:1
Total events: 905754
Landau-Gauss fit, trigg.
L MPV = 45.95 ± 0.18
Max = 50.74 ± 0.19
χ²/ndf = 20.20
SNR = 2.01
r:1 c:1 m:1
Total events: 901080
Landau-Gauss fit, trigg.
L MPV = 47.52 ± 0.18
Max = 54.60 ± 0.21
χ²/ndf = 36.85
SNR = 2.04
r:1 c:2 m:1
Total events: 895787
Landau-Gauss fit, trigg.
L MPV = 40.68 ± 0.14
Max = 47.43 ± 0.17
χ²/ndf = 43.03
SNR = 1.54
r:1 c:3 m:1 layer 4
Total events: 880652
Landau-Gauss fit, trigg.
L MPV = 49.39 ± 0.18
Max = 57.29 ± 0.21
χ²/ndf = 52.89
SNR = 1.70
μ⁻-beam, E_b = 5 GeV
Run 86, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 888590
Landau-Gauss fit, trigg.
L MPV = 55.28 ± 0.23
Max = 60.68 ± 0.25
χ²/ndf = 22.01
SNR = 2.00
r:0 c:1 m:1
Total events: 902889
Landau-Gauss fit, trigg.
L MPV = 43.59 ± 0.15
Max = 49.63 ± 0.17
χ²/ndf = 33.09
SNR = 1.58
r:0 c:2 m:1
Total events: 879888
Landau-Gauss fit, trigg.
L MPV = 45.98 ± 0.20
Max = 54.19 ± 0.24
χ²/ndf = 56.76
SNR = 1.40
r:0 c:3 m:1
Total events: 869169
Landau-Gauss fit, trigg.
L MPV = 54.17 ± 0.16
Max = 59.62 ± 0.18
χ²/ndf = 23.80
SNR = 1.46
r:1 c:0 m:0
Total events: 1099894
Landau-Gauss fit, trigg.
L MPV = 47.39 ± 0.23
Max = 53.21 ± 0.26
χ²/ndf = 18.35
SNR = 1.53
r:1 c:1 m:0
Total events: 1096158
Landau-Gauss fit, trigg.
L MPV = 44.19 ± 0.20
Max = 51.73 ± 0.23
χ²/ndf = 41.11
SNR = 1.54
r:1 c:2 m:0
Total events: 1093037
Landau-Gauss fit, trigg.
L MPV = 43.03 ± 0.15
Max = 50.19 ± 0.18
χ²/ndf = 54.20
SNR = 1.56
r:1 c:3 m:0
Total events: 1090835
Landau-Gauss fit, trigg.
L MPV = 46.72 ± 0.17
Max = 54.31 ± 0.20
χ²/ndf = 57.18
SNR = 1.57
r:0 c:0 m:0
Total events: 1098745
Landau-Gauss fit, trigg.
L MPV = 50.74 ± 0.15
Max = 53.94 ± 0.16
χ²/ndf = 9.83
SNR = 1.92
r:0 c:1 m:0
Total events: 1098127
Landau-Gauss fit, trigg.
L MPV = 55.77 ± 0.14
Max = 60.32 ± 0.15
χ²/ndf = 13.07
SNR = 1.80
r:0 c:2 m:0
Total events: 1094274
Landau-Gauss fit, trigg.
L MPV = 46.64 ± 0.18
Max = 53.84 ± 0.21
χ²/ndf = 41.47
SNR = 2.02
r:0 c:3 m:0
Total events: 1093895
Landau-Gauss fit, trigg.
L MPV = 49.57 ± 0.19
Max = 55.65 ± 0.21
χ²/ndf = 23.86
SNR = 1.57
counts
ADC (arb. units)

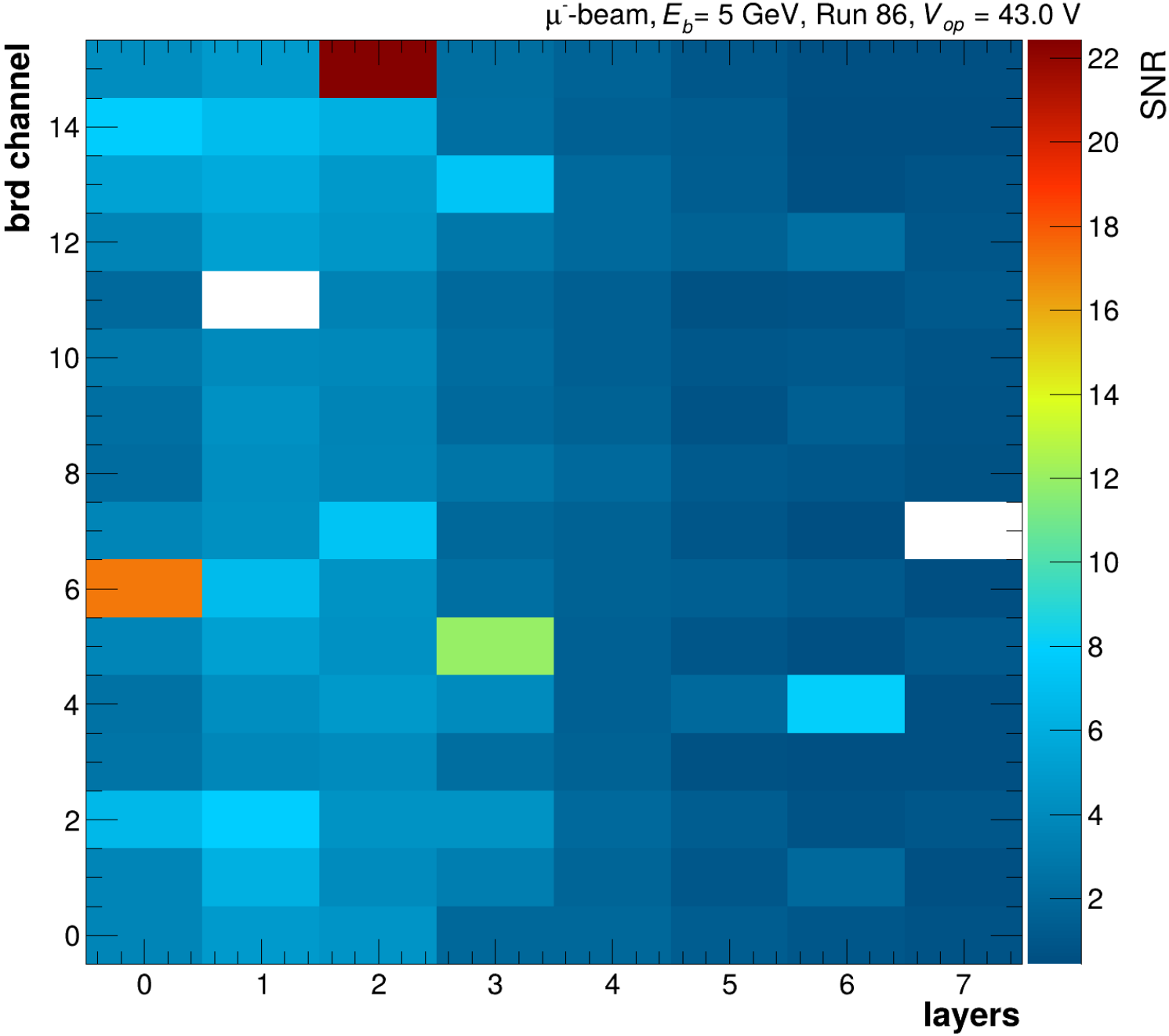

r:1 c:0 m:1
Total events: 892784
Landau-Gauss fit, trigg.
L MPV = 54.33 ± 0.18
Max = 58.27 ± 0.19
χ²/ndf = 11.30
SNR = 1.62
r:1 c:1 m:1
Total events: 892808
Landau-Gauss fit, trigg.
L MPV = 53.19 ± 0.12
Max = 57.03 ± 0.13
χ²/ndf = 19.47
SNR = 1.36
r:1 c:2 m:1
Total events: 844086
Landau-Gauss fit, trigg.
L MPV = 60.97 ± 0.15
Max = 66.99 ± 0.16
χ²/ndf = 33.19
SNR = 1.29
r:1 c:3 m:1 layer 5
Total events: 884747
Landau-Gauss fit, trigg.
L MPV = 51.19 ± 0.19
Max = 58.15 ± 0.21
χ²/ndf = 43.18
SNR = 1.01
μ⁻-beam, E_b = 5 GeV
Run 86, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 906565
Landau-Gauss fit, trigg.
L MPV = 54.20 ± 0.18
Max = 58.51 ± 0.20
χ²/ndf = 12.01
SNR = 1.20
r:0 c:1 m:1
Total events: 890069
Landau-Gauss fit, trigg.
L MPV = 54.15 ± 0.11
Max = 57.98 ± 0.12
χ²/ndf = 27.10
SNR = 0.72
r:0 c:2 m:1
Total events: 881702
Landau-Gauss fit, trigg.
L MPV = 51.19 ± 0.24
Max = 58.80 ± 0.28
χ²/ndf = 50.14
SNR = 1.02
r:0 c:3 m:1
Total events: 898544
Landau-Gauss fit, trigg.
L MPV = 55.28 ± 0.18
Max = 61.03 ± 0.20
χ²/ndf = 14.58
SNR = 0.63
r:1 c:0 m:0
Total events: 1098748
Landau-Gauss fit, trigg.
L MPV = 58.90 ± 0.24
Max = 64.12 ± 0.26
χ²/ndf = 15.74
SNR = 2.02
r:1 c:1 m:0
Total events: 1097045
Landau-Gauss fit, trigg.
L MPV = 59.94 ± 0.18
Max = 66.18 ± 0.20
χ²/ndf = 30.77
SNR = 0.93
r:1 c:2 m:0
Total events: 1085897
Landau-Gauss fit, trigg.
L MPV = 49.82 ± 0.13
Max = 55.43 ± 0.15
χ²/ndf = 38.22
SNR = 1.42
r:1 c:3 m:0
Total events: 1095320
Landau-Gauss fit, trigg.
L MPV = 50.98 ± 0.21
Max = 58.68 ± 0.24
χ²/ndf = 39.29
SNR = 0.91
r:0 c:0 m:0
Total events: 1100826
Landau-Gauss fit, trigg.
L MPV = 58.42 ± 0.29
Max = 63.78 ± 0.31
χ²/ndf = 11.76
SNR = 1.34
r:0 c:1 m:0
Total events: 1098149
Landau-Gauss fit, trigg.
L MPV = 58.35 ± 0.18
Max = 64.00 ± 0.20
χ²/ndf = 35.84
SNR = 1.03
r:0 c:2 m:0
Total events: 1096641
Landau-Gauss fit, trigg.
L MPV = 53.45 ± 0.21
Max = 60.36 ± 0.24
χ²/ndf = 34.32
SNR = 1.36
r:0 c:3 m:0
Total events: 1099034
Landau-Gauss fit, trigg.
L MPV = 52.20 ± 0.29
Max = 59.46 ± 0.33
χ²/ndf = 25.23
SNR = 0.68
counts
ADC (arb. units)

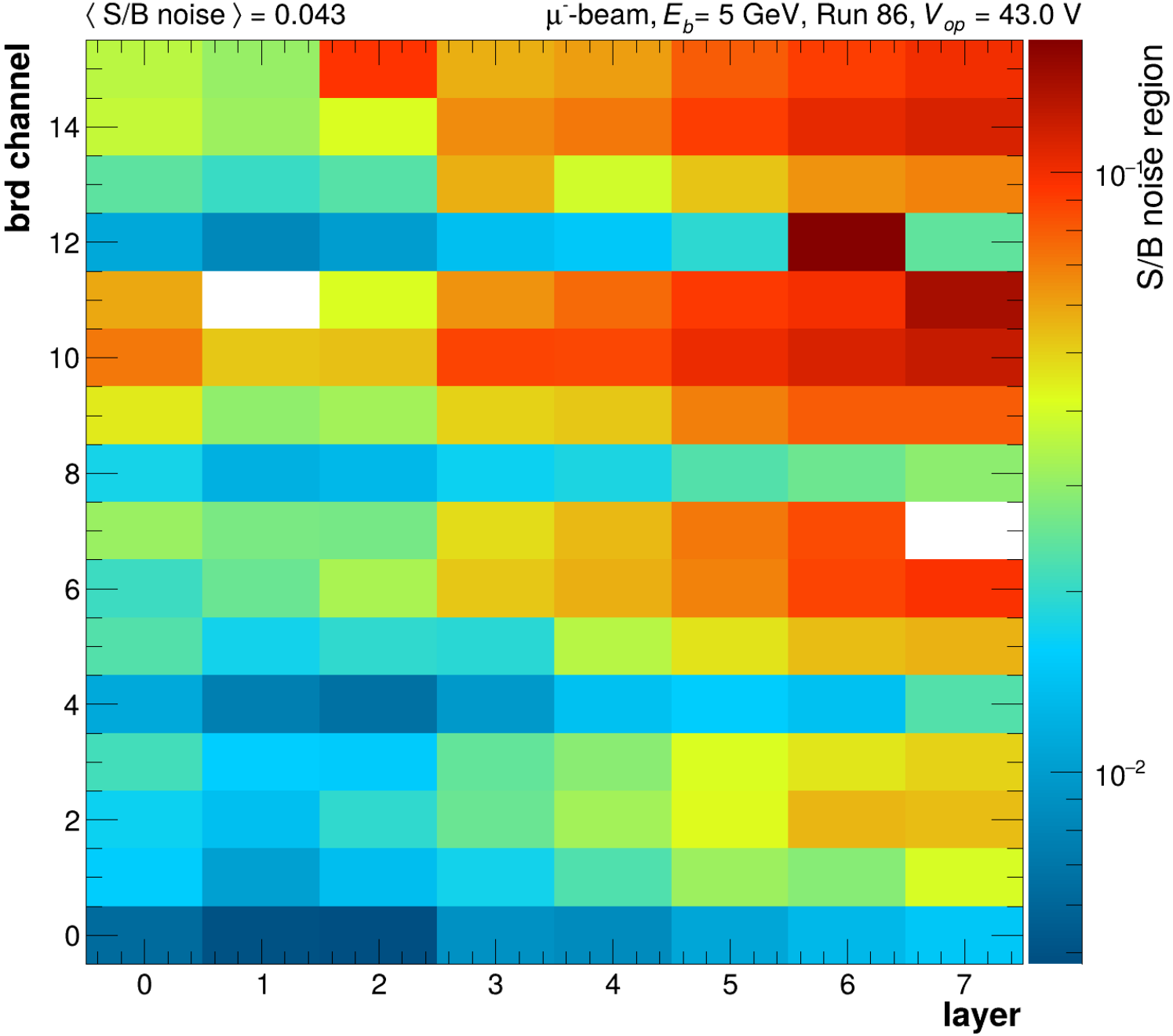

r:1 c:0 m:1
Total events: 185933
Landau-Gauss fit, trigg.
L MPV = 69.84 ± 0.37
Max = 75.17 ± 0.39
χ²/ndf = 22.80
SNR = 2.41
r:1 c:1 m:1
Total events: 898542
Landau-Gauss fit, trigg.
L MPV = 66.80 ± 0.20
Max = 72.43 ± 0.22
χ²/ndf = 14.41
SNR = 0.59
r:1 c:2 m:1
Total events: 888414
Landau-Gauss fit, trigg.
L MPV = 60.77 ± 0.18
Max = 66.76 ± 0.20
χ²/ndf = 44.20
SNR = 0.50
r:1 c:3 m:1 layer 6
Total events: 898527
Landau-Gauss fit, trigg.
L MPV = 69.02 ± 0.21
Max = 75.48 ± 0.23
χ²/ndf = 18.11
SNR = 0.61
μ⁻-beam, E_b = 5 GeV
Run 86, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 906106
Landau-Gauss fit, trigg.
L MPV = 57.22 ± 0.39
Max = 63.23 ± 0.43
χ²/ndf = 13.62
SNR = 1.01
r:0 c:1 m:1
Total events: 750776
Landau-Gauss fit, trigg.
L MPV = 67.20 ± 0.31
Max = 73.60 ± 0.34
χ²/ndf = 43.88
SNR = 1.42
r:0 c:2 m:1
Total events: 892229
Landau-Gauss fit, trigg.
L MPV = 65.02 ± 0.17
Max = 70.69 ± 0.19
χ²/ndf = 23.56
SNR = 1.09
r:0 c:3 m:1
Total events: 898570
Landau-Gauss fit, trigg.
L MPV = 65.34 ± 0.22
Max = 71.31 ± 0.24
χ²/ndf = 12.35
SNR = 0.70
r:1 c:0 m:0
Total events: 932944
Landau-Gauss fit, trigg.
L MPV = 60.58 ± 0.32
Max = 65.97 ± 0.35
χ²/ndf = 19.08
SNR = 8.02
r:1 c:1 m:0
Total events: 1099554
Landau-Gauss fit, trigg.
L MPV = 66.61 ± 0.19
Max = 72.33 ± 0.21
χ²/ndf = 18.64
SNR = 0.59
r:1 c:2 m:0
Total events: 1097420
Landau-Gauss fit, trigg.
L MPV = 64.36 ± 0.20
Max = 70.97 ± 0.22
χ²/ndf = 50.14
SNR = 1.12
r:1 c:3 m:0
Total events: 1099285
Landau-Gauss fit, trigg.
L MPV = 53.07 ± 0.30
Max = 60.17 ± 0.34
χ²/ndf = 37.22
SNR = 0.50
r:0 c:0 m:0
Total events: 1100991
Landau-Gauss fit, trigg.
L MPV = 61.66 ± 0.34
Max = 67.03 ± 0.37
χ²/ndf = 7.80
SNR = 0.99
r:0 c:1 m:0
Total events: 1060663
Landau-Gauss fit, trigg.
L MPV = 59.06 ± 0.23
Max = 64.69 ± 0.25
χ²/ndf = 32.98
SNR = 1.99
r:0 c:2 m:0
Total events: 1098445
Landau-Gauss fit, trigg.
L MPV = 63.47 ± 0.17
Max = 68.78 ± 0.19
χ²/ndf = 17.23
SNR = 0.71
r:0 c:3 m:0
Total events: 1099911
Landau-Gauss fit, trigg.
L MPV = 61.54 ± 0.28
Max = 67.92 ± 0.31
χ²/ndf = 16.73
SNR = 0.60
counts
ADC (arb. units)

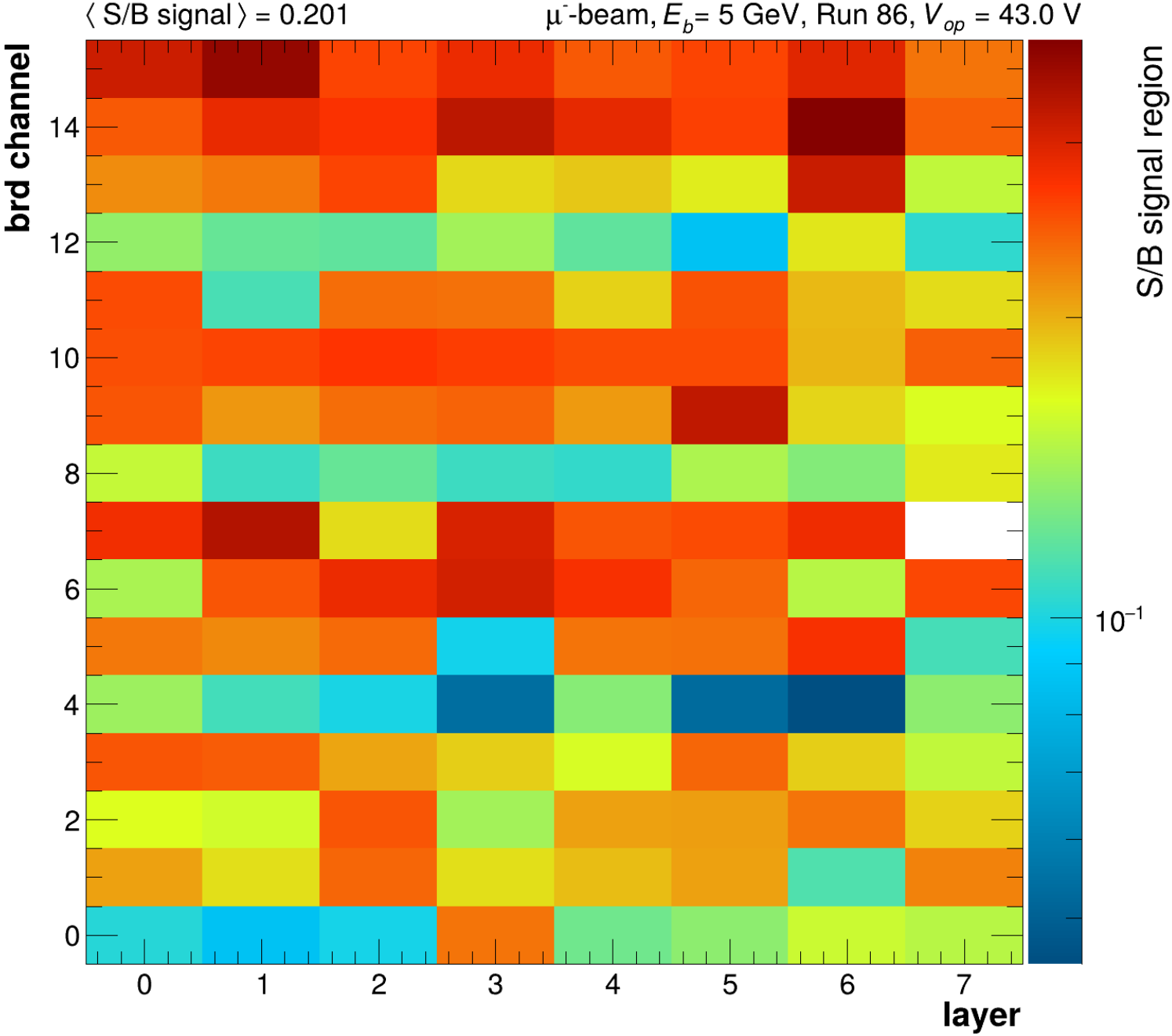

r:1 c:0 m:1
Total events: 906533
Landau-Gauss fit, trigg.
L MPV = 53.08 ± 0.32
Max = 52.83 ± 0.32
χ²/ndf = 18.56
SNR = 0.95
r:1 c:1 m:1
Total events: 892945
Landau-Gauss fit, trigg.
L MPV = 70.35 ± 0.31
Max = 76.36 ± 0.33
χ²/ndf = 20.12
SNR = 0.80
r:1 c:2 m:1
Total events: 889245
Landau-Gauss fit, trigg.
L MPV = 68.88 ± 0.20
Max = 74.55 ± 0.21
χ²/ndf = 29.29
SNR = 0.52
r:1 c:3 m:1 layer 7
Total events: 882585
Landau-Gauss fit, trigg.
L MPV = 62.13 ± 0.27
Max = 64.68 ± 0.28
χ²/ndf = 28.51
SNR = 0.55
μ⁻-beam, E_b = 5 GeV
Run 86, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 900664
Landau-Gauss fit, trigg.
L MPV = 45.60 ± 0.12
Max = 47.08 ± 0.12
χ²/ndf = 31.34
SNR = 0.68
r:0 c:1 m:1
Total events: 892477
Landau-Gauss fit, trigg.
L MPV = 73.31 ± 0.32
Max = 79.62 ± 0.35
χ²/ndf = 25.73
SNR = 0.75
r:0 c:2 m:1
Total events: 846467
Landau-Gauss fit, trigg.
L MPV = 65.08 ± 0.37
Max = 71.60 ± 0.41
χ²/ndf = 37.57
SNR = 0.83
r:0 c:3 m:1
Total events: 676996
Landau-Gauss fit, trigg.
L MPV = 73.50 ± 0.40
Max = 80.52 ± 0.44
χ²/ndf = 24.78
SNR = 1.05
r:1 c:0 m:0
Total events: 1101044
Landau-Gauss fit, trigg.
L MPV = 63.16 ± 0.42
Max = 67.98 ± 0.46
χ²/ndf = 12.40
SNR = 0.60
r:1 c:1 m:0
Total events: 1098939
Landau-Gauss fit, trigg.
L MPV = 70.75 ± 0.30
Max = 77.24 ± 0.32
χ²/ndf = 22.89
SNR = 1.07
r:1 c:2 m:0
Total events: 1098513
Landau-Gauss fit, trigg.
L MPV = 71.48 ± 0.28
Max = 78.14 ± 0.30
χ²/ndf = 38.54
SNR = 0.44
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 1100865
Landau-Gauss fit, trigg.
L MPV = 63.22 ± 0.42
Max = 62.89 ± 0.42
χ²/ndf = 13.56
SNR = 0.76
r:0 c:1 m:0
Total events: 1099133
Landau-Gauss fit, trigg.
L MPV = 70.93 ± 0.27
Max = 76.80 ± 0.29
χ²/ndf = 11.90
SNR = 0.56
r:0 c:2 m:0
Total events: 1088484
Landau-Gauss fit, trigg.
L MPV = 68.77 ± 0.31
Max = 75.59 ± 0.34
χ²/ndf = 23.50
SNR = 0.97
r:0 c:3 m:0
Total events: 1042420
Landau-Gauss fit, trigg.
L MPV = 68.02 ± 0.32
Max = 74.19 ± 0.35
χ²/ndf = 15.88
SNR = 0.59
counts
ADC (arb. units)

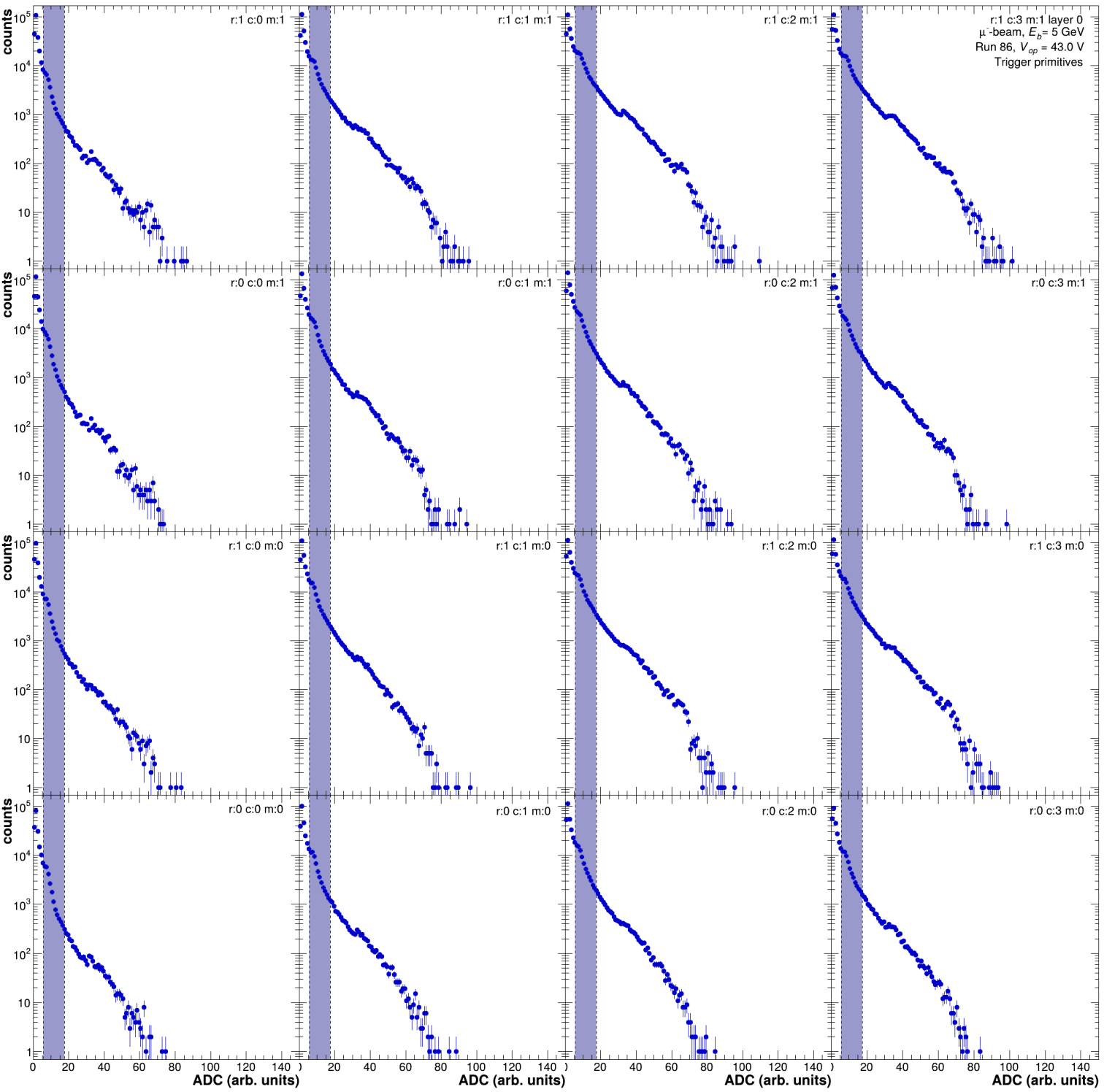

μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
1000
900
800
700
600
500
400
300

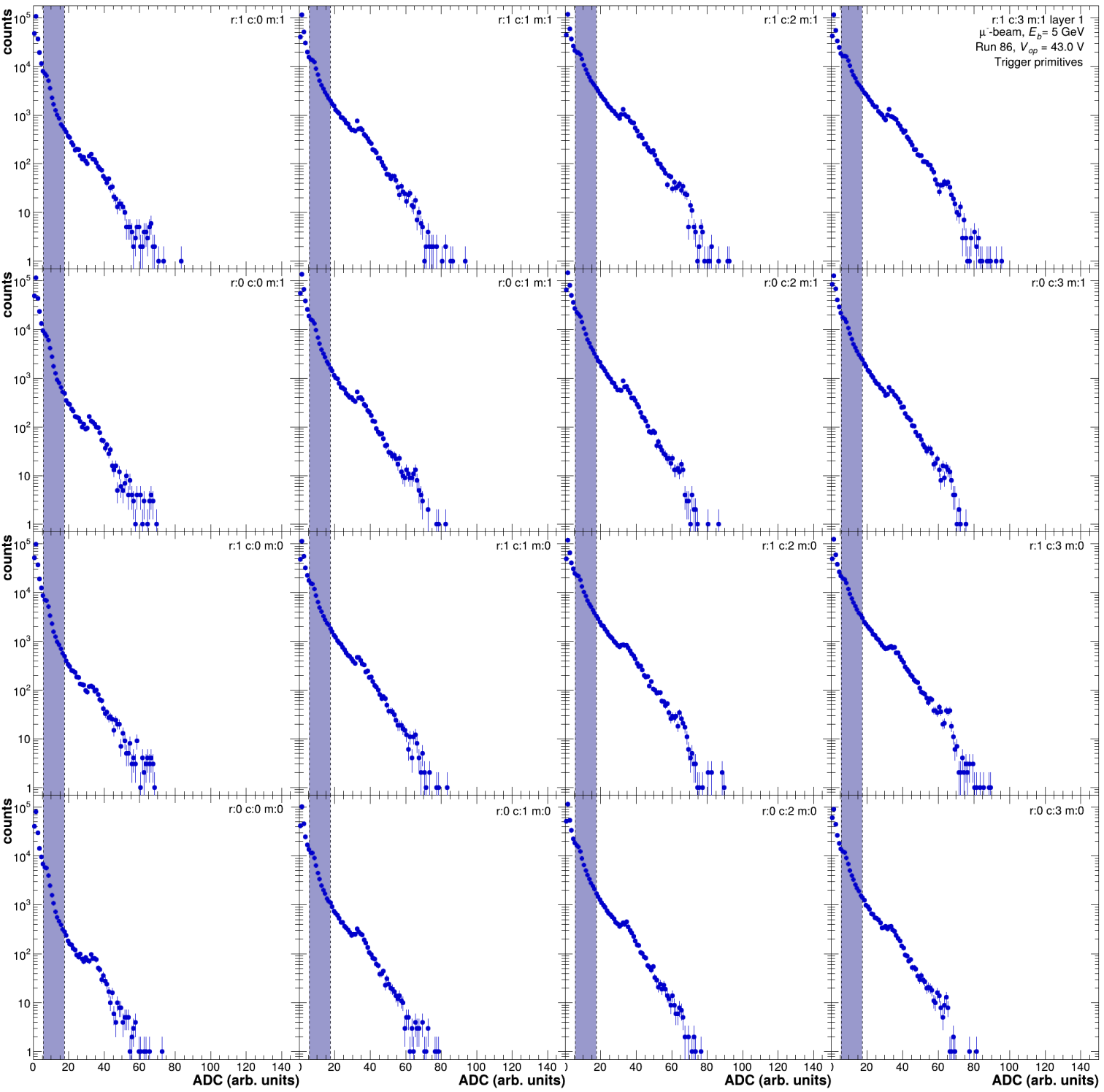

evt. collected = 1101322
μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
layer
counts
10^5

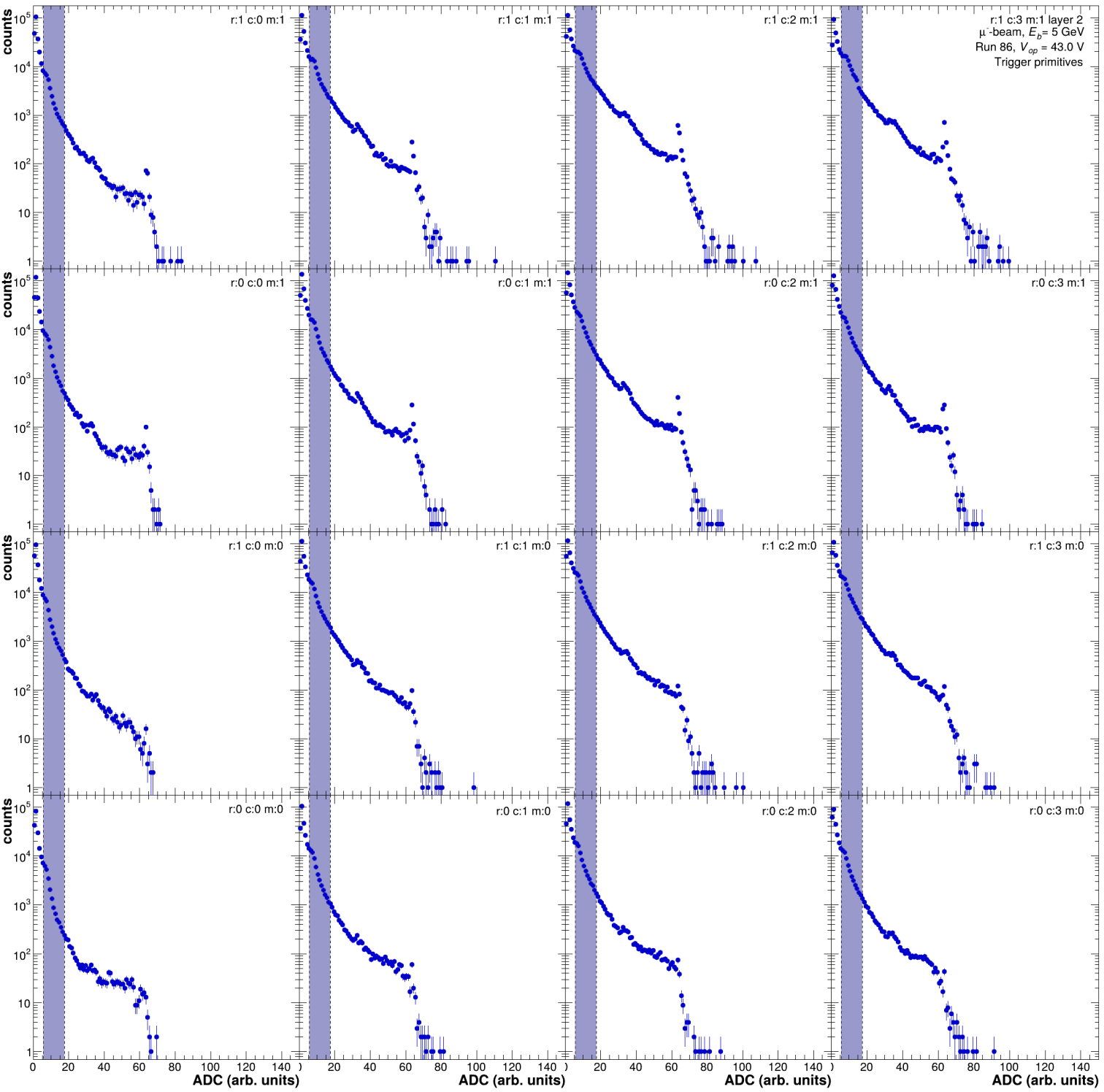

μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
2
4
6
8
10
12
14
16
18
20
22

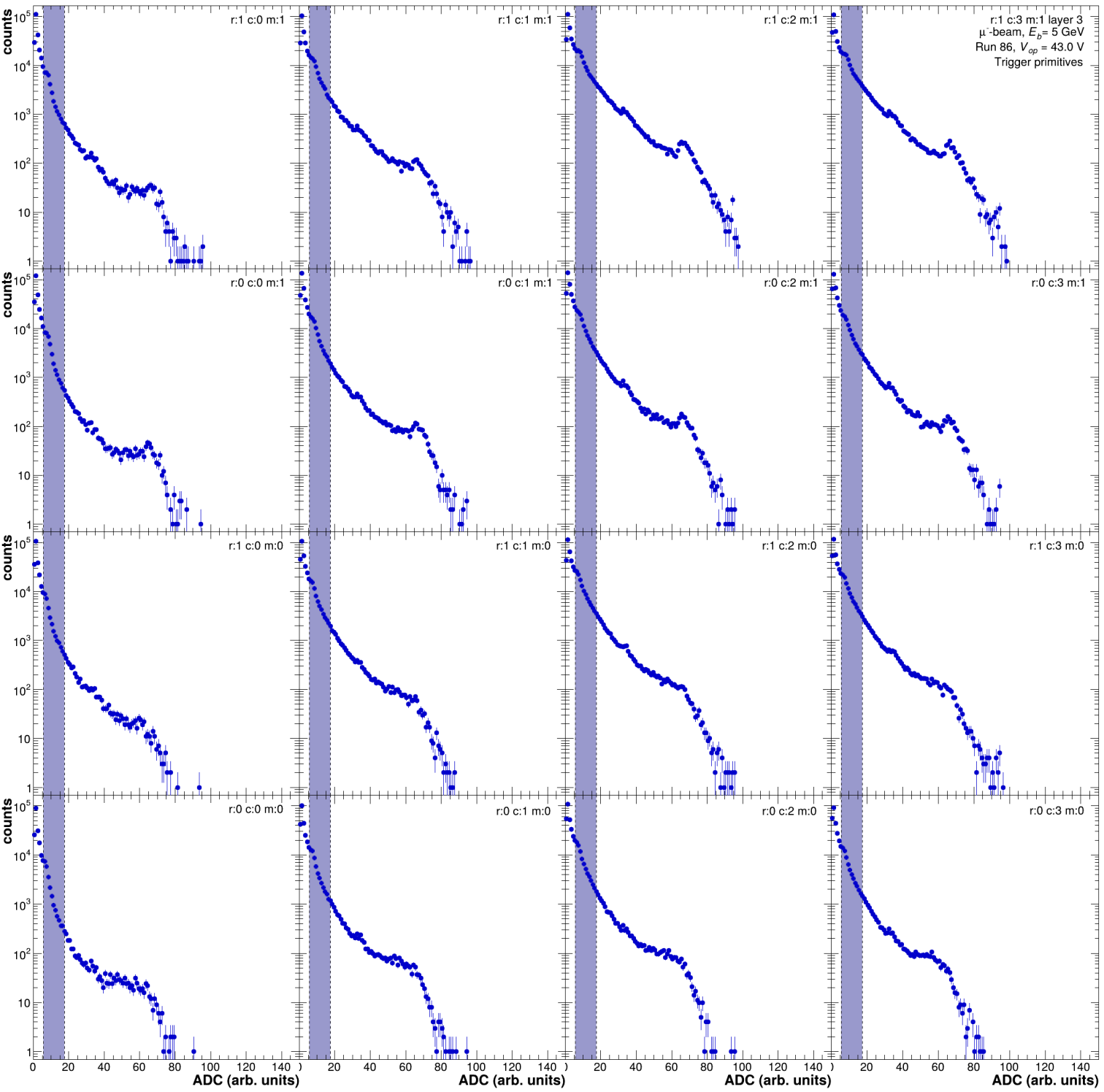

⟨ S/B noise ⟩ = 0.043
μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

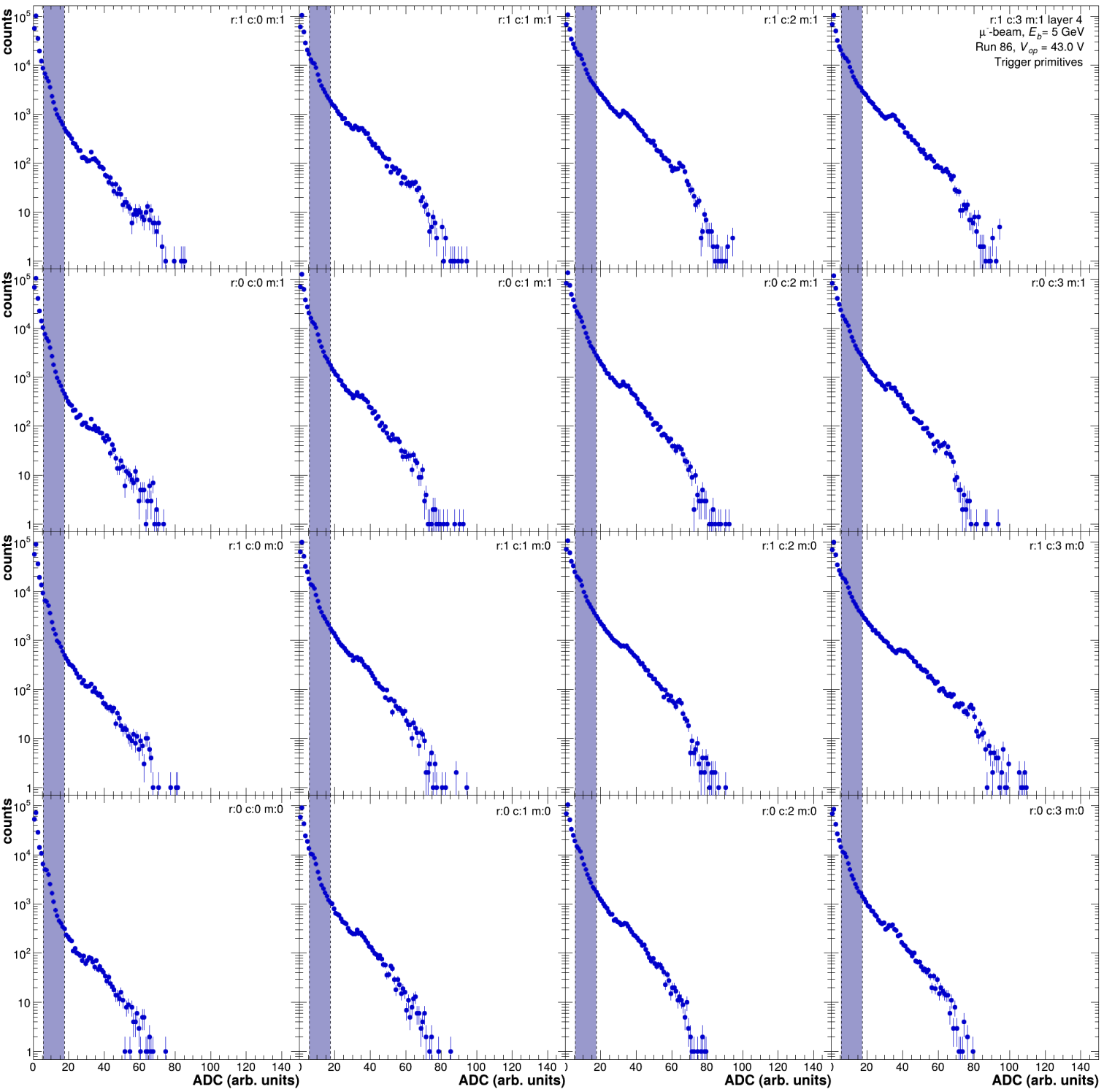

⟨ S/B signal ⟩ = 0.201
μ⁻-beam, E_b= 5 GeV, Run 86, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

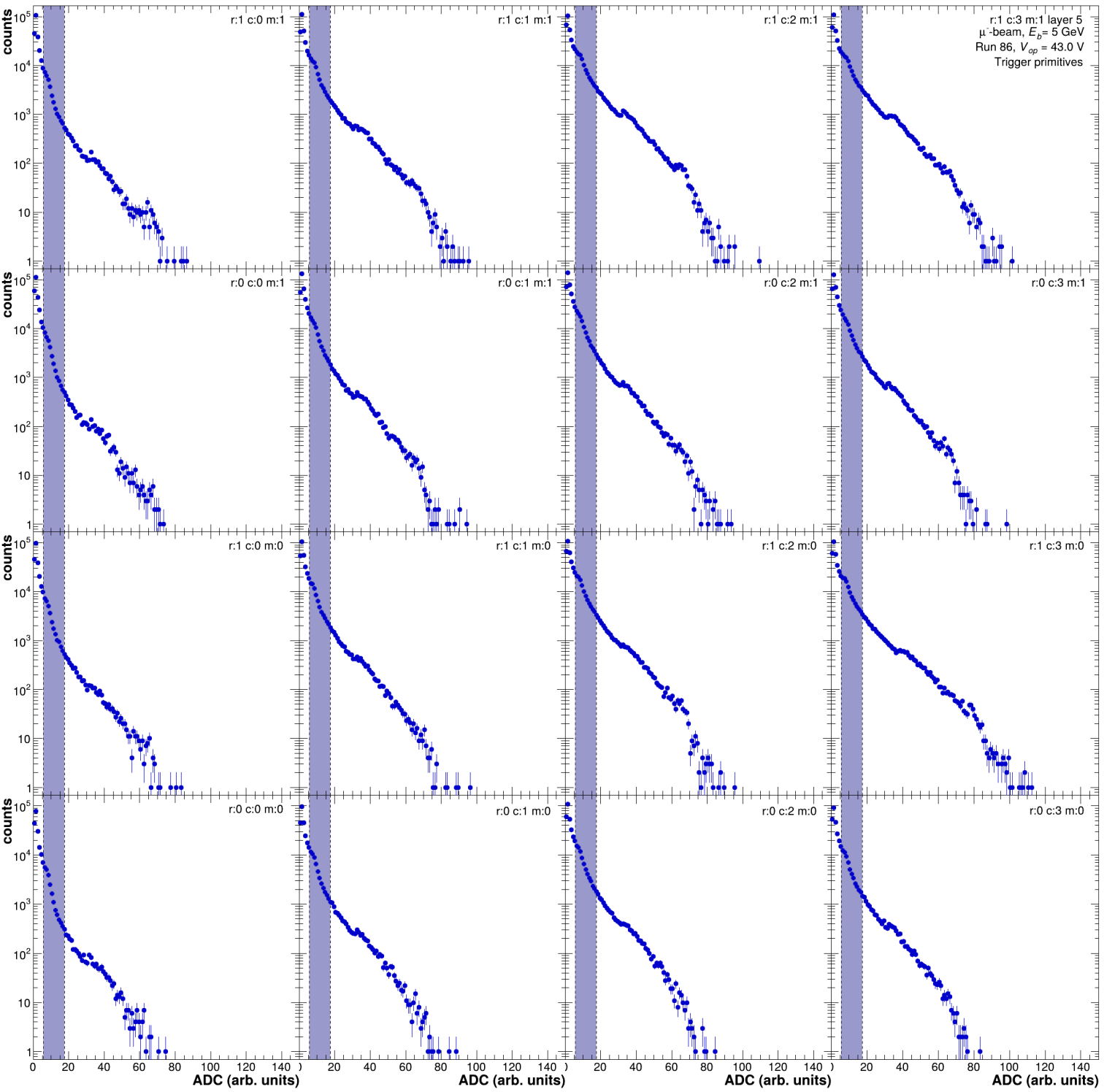

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

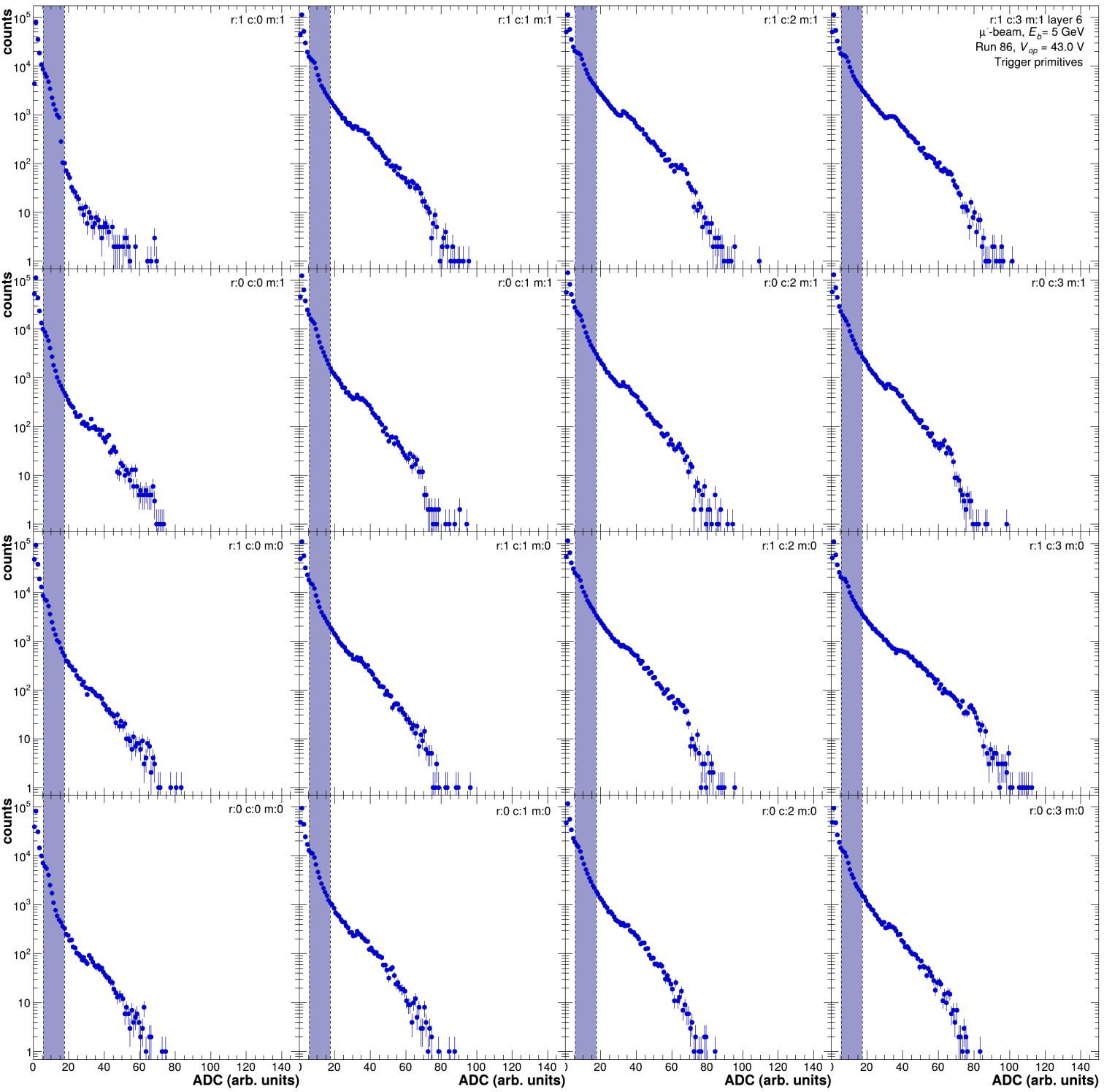

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

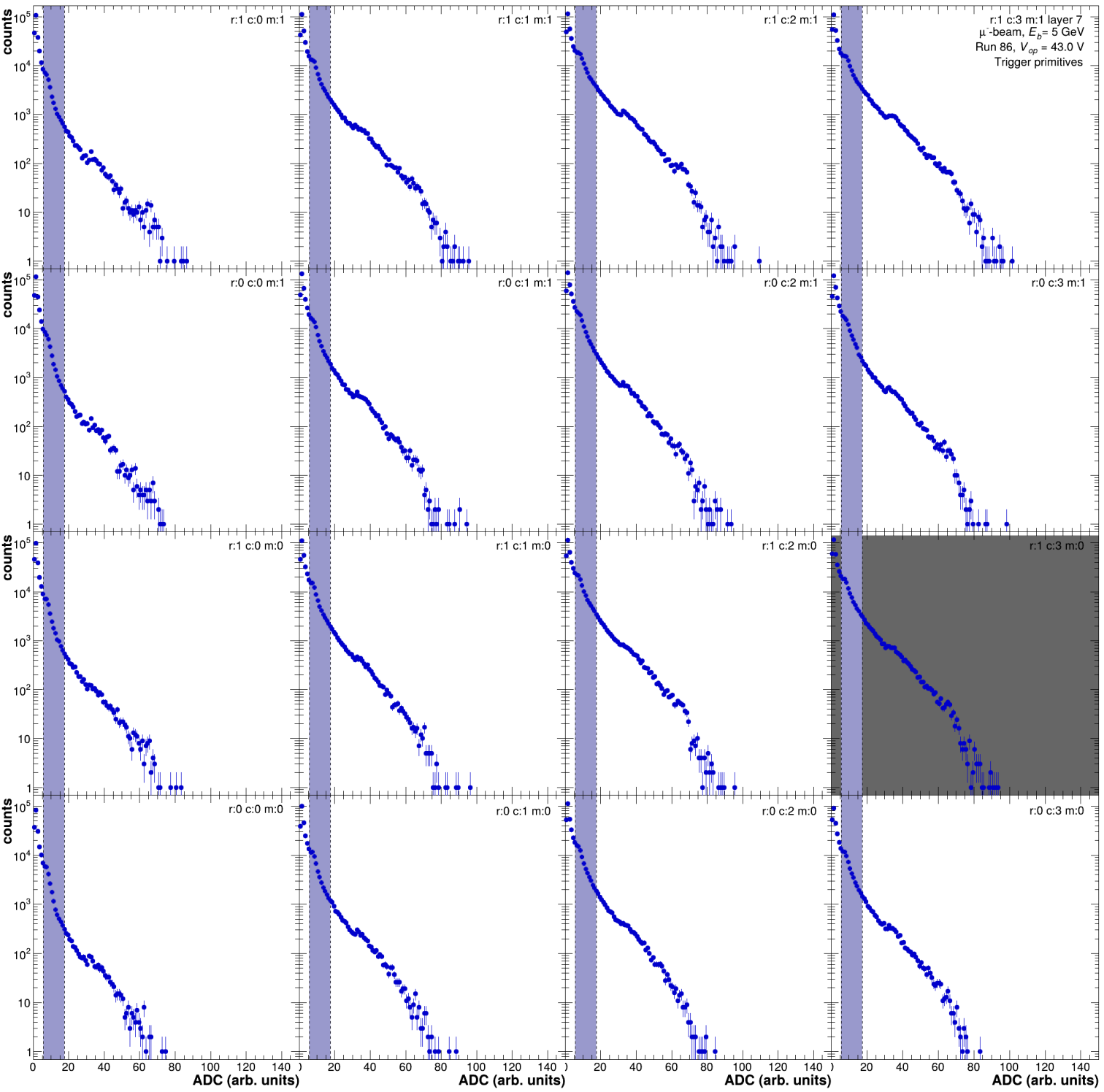

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 86, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)